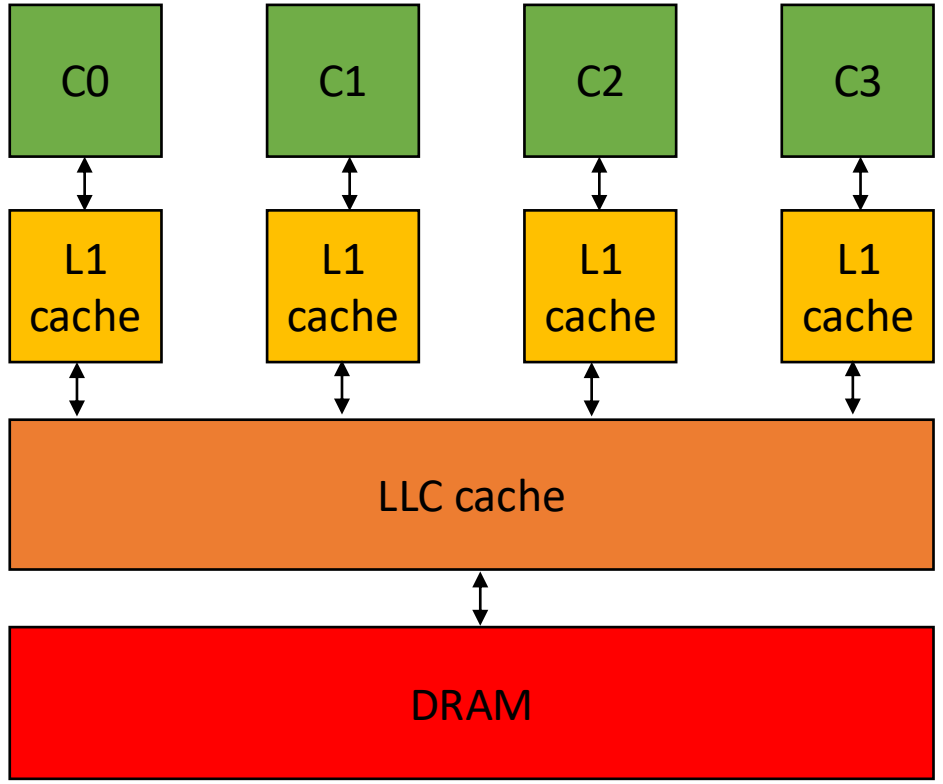
CSE113: Parallel Programming

- **Topic**: Architecture and Compiler Overview
  - Programming Language to ISA compilation
  - 3-address code
  - multiprocessors
  - memory hierarchy



Announcements

- Homework 1 released tonight!
  - A week to do it
  - 3 free late days

- Solutions require a design doc.
  - Not harshly graded but liable to lose points for low-effort
  - Forces you to think about your solution before you start

Announcements

- Instructor and TAs: Office hours announced on the webpage
- Tutors: Office hours one announced, another announced by Thursday.

Review

In a perfect world...

- Historically this worked well



The negotiators:
Specifications
Compiles
Runtimes
Interpreters

- Dennard's scaling:
  - Computer speed doubles every 1.5 years.

2003

700 MHz



3x ***increase***
over 4 years

2007

2.1 GHz



However…

These trends slowed down in ~2007



The negotiators:
Specifications
Compiles
Runtimes
Interpreters

2007
2.1 GHz



2 cores

1.2x increase
over 10 years

2017
2.5 GHz



4 cores

Python?

- Great language for scripting
  - We will use it to automate experiments in this class
- The GIL (global interpreter lock) restricts parallelism significantly.
  - Makes the language safe
- TensorFlow and Pytorch?
  - Wrappers around low-level kernels that execute outside of the python interpreter

Lecture Schedule

- Overview - why do we need a lecture on compilation and architecture?

- **Compilation** - *How do we translate a program from a human-accessible language to a language that the processor understands?*

- Architecture - How do processors execute programs?

- Example

Compilation:

Language



ISA



Compilation:

Language



ISA




```
int add(int a, int b) {
    return a + b;
}
```

Compilation:

Language



ISA




```
int add(int a, int b) {
    return a + b;
}
```

If we didn't have computers, would this mean anything?

Compilation:

Language



ISA




```
int add(int a, int b) {
    return a + b;
}
```

Officially defined by the specification

ISO standard: costs $200

~1400 pages

Compilation:

Language




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int add(int a, int b) {
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ISA



Official specification
Intel provides a specification: *free*
2200 pages

Compilation:

Language




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int add(int a, int b) {
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ISA




```
???
```

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Compilation:

Language




```
int add(int a, int b) {
    return a + b;
}
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~1400 pages




```
add(int, int): # @add(int, int)
push rbp
mov rbp, rsp
mov dword ptr [rbp - 4], edi
mov dword ptr [rbp - 8], esi
mov eax, dword ptr [rbp - 4]
add eax, dword ptr [rbp - 8]
pop rbp
ret
```

Official specification

Intel provides a specification: *free*

2200 pages

Compilation:

Language




```
int add(int a, int b) {
    return a + b;
}
```




```
add(int, int):
sub sp, sp, #16
str w0, [sp, #12]
str w1, [sp, #8]
ldr w8, [sp, #12]
ldr w9, [sp, #8]
add w0, w8, w9
add sp, sp, #16
ret
```

Officially defined by the specification

ISO standard: costs $200

~1400 pages

How about a more complicated program?

Quadratic formula

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Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

How about a more complicated program?

Quadratic formula

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```
x = (-b - sqrt(b*b - 4 * a * c)) / (2*a)
```

How about a more complicated program?

Quadratic formula

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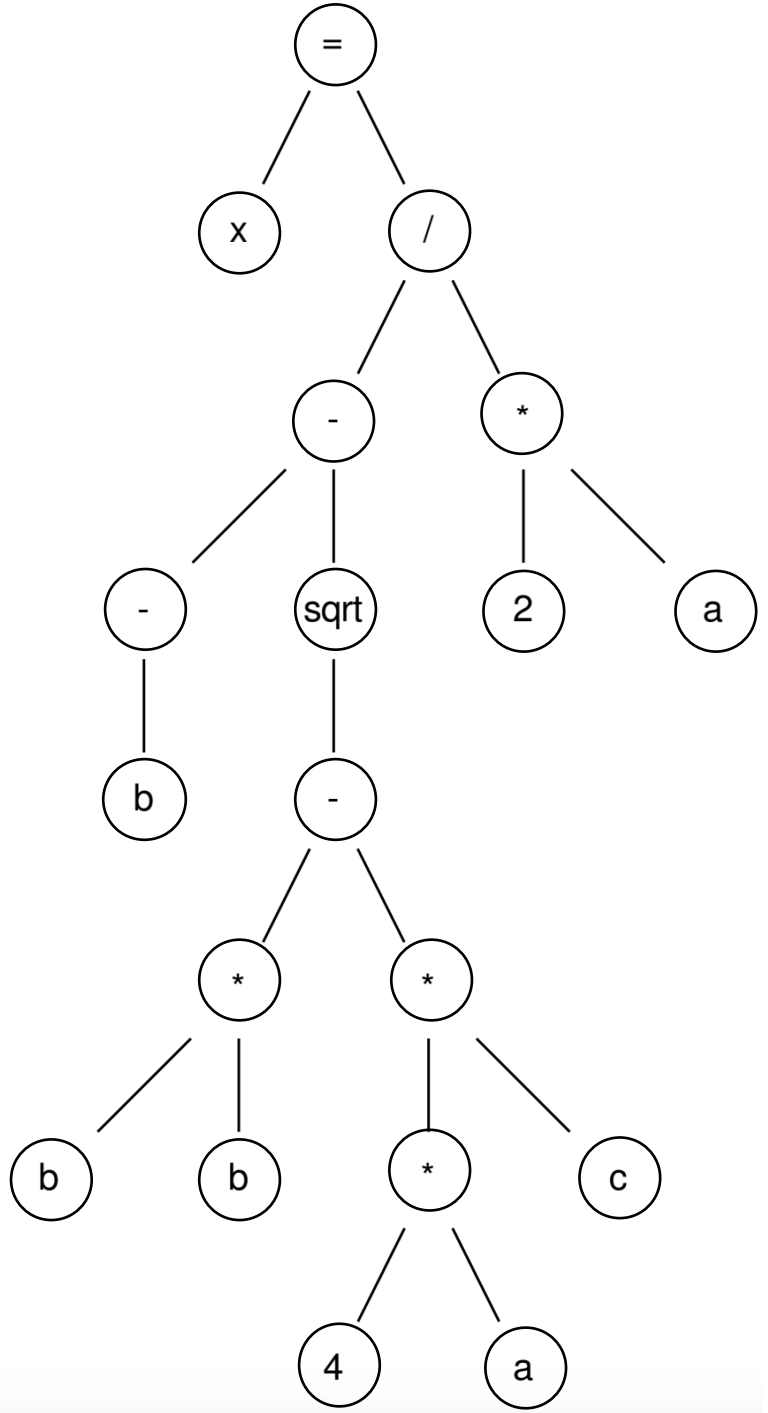


Official specification
Intel provides a specification: *free*
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There is not an ISA instruction that combines all these instructions!

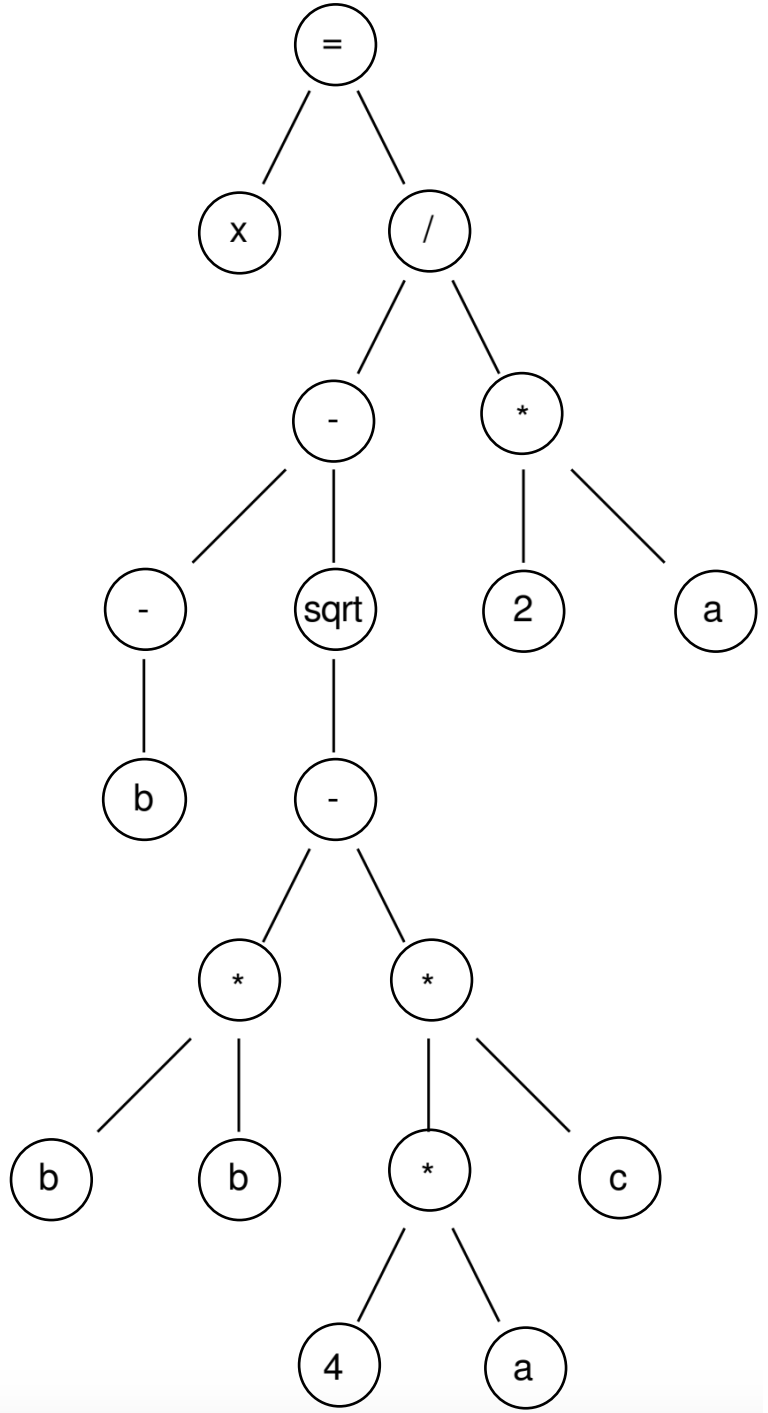
```
x = (-b - sqrt(b*b - 4 * a * c)) / (2*a)
```

A compiler will turn this into an *abstract syntax tree* (AST)



Simplify this code:

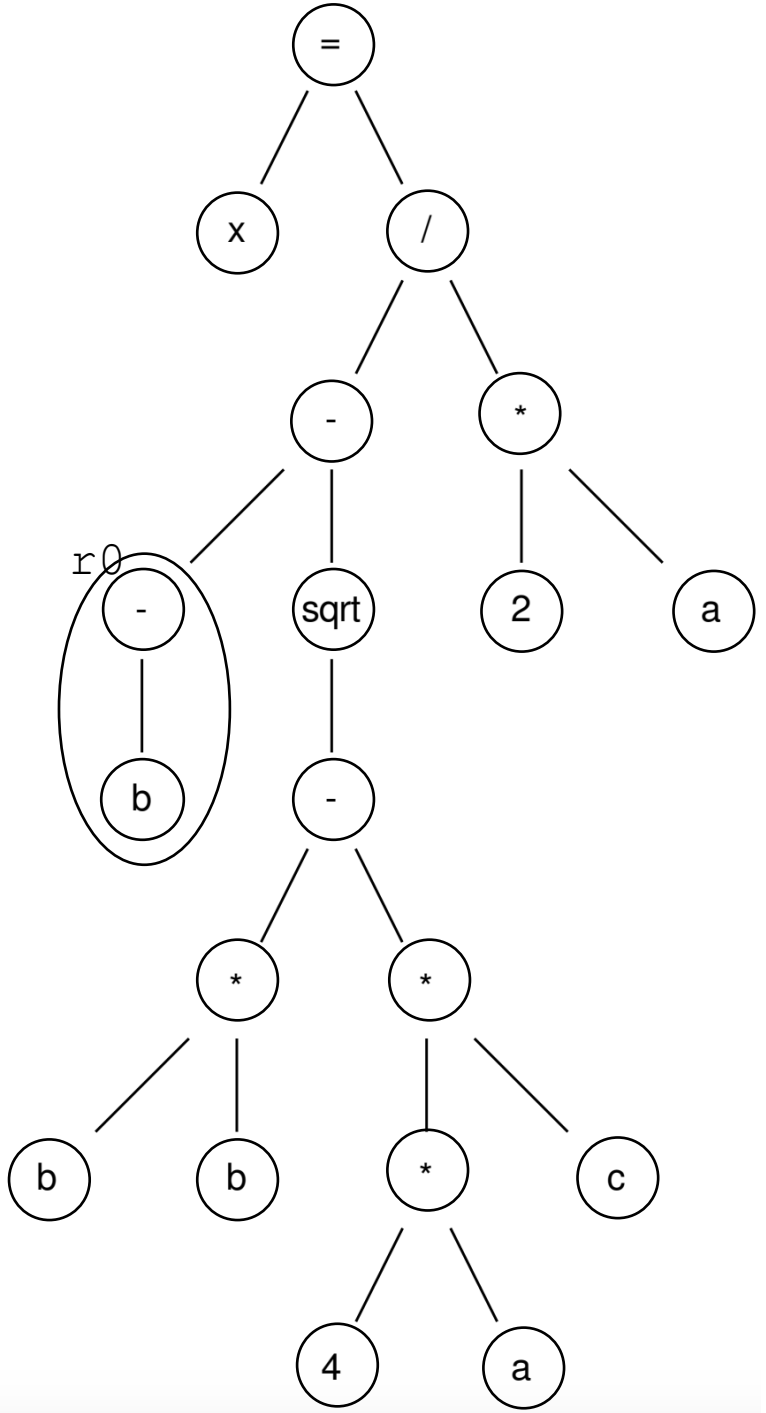
Post-order traversal, using temporary variables



Simplify this code:

post-order traversal, using temporary variables

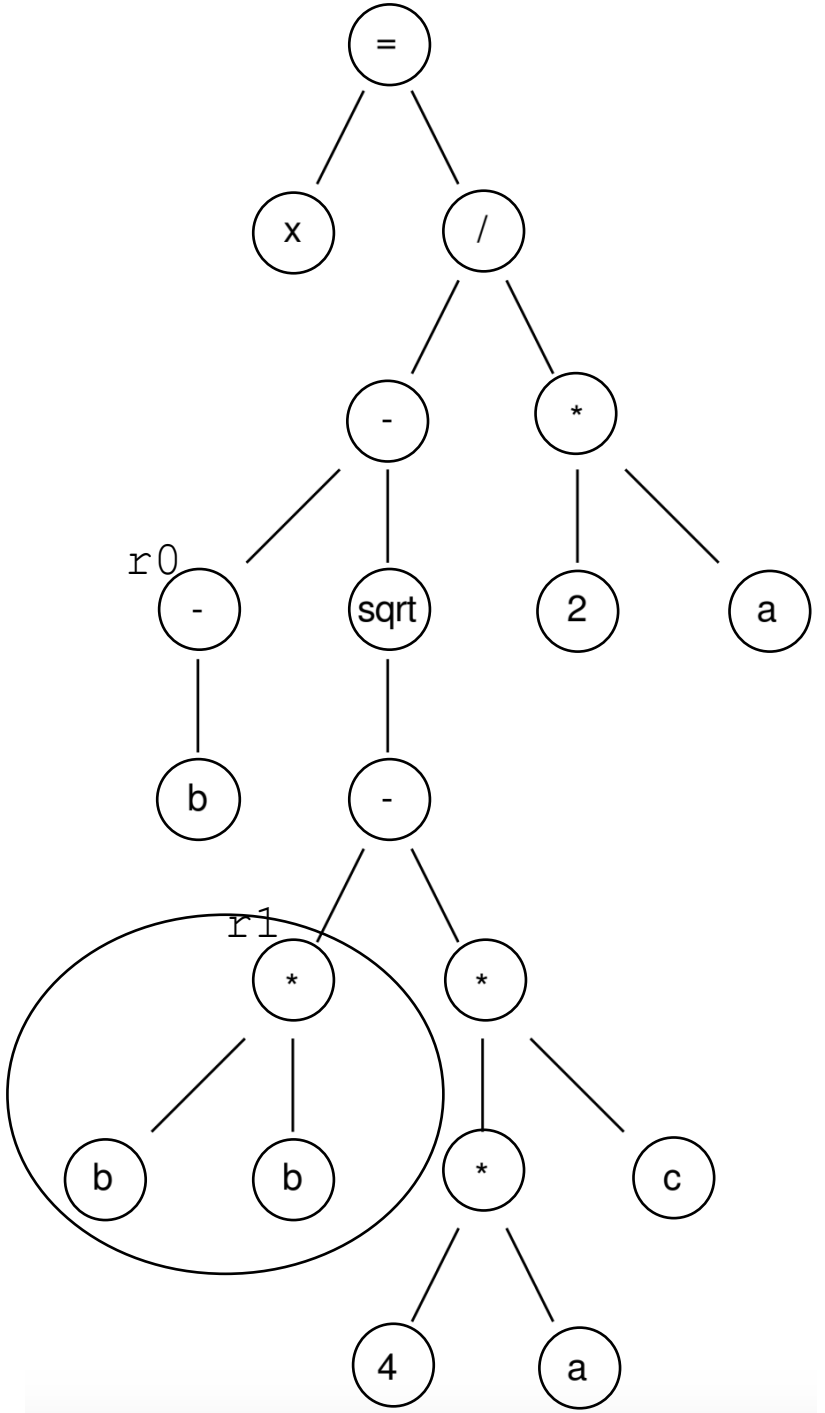
```
r0 = neg(b);
```



Simplify this code:

post-order traversal, using temporary variables

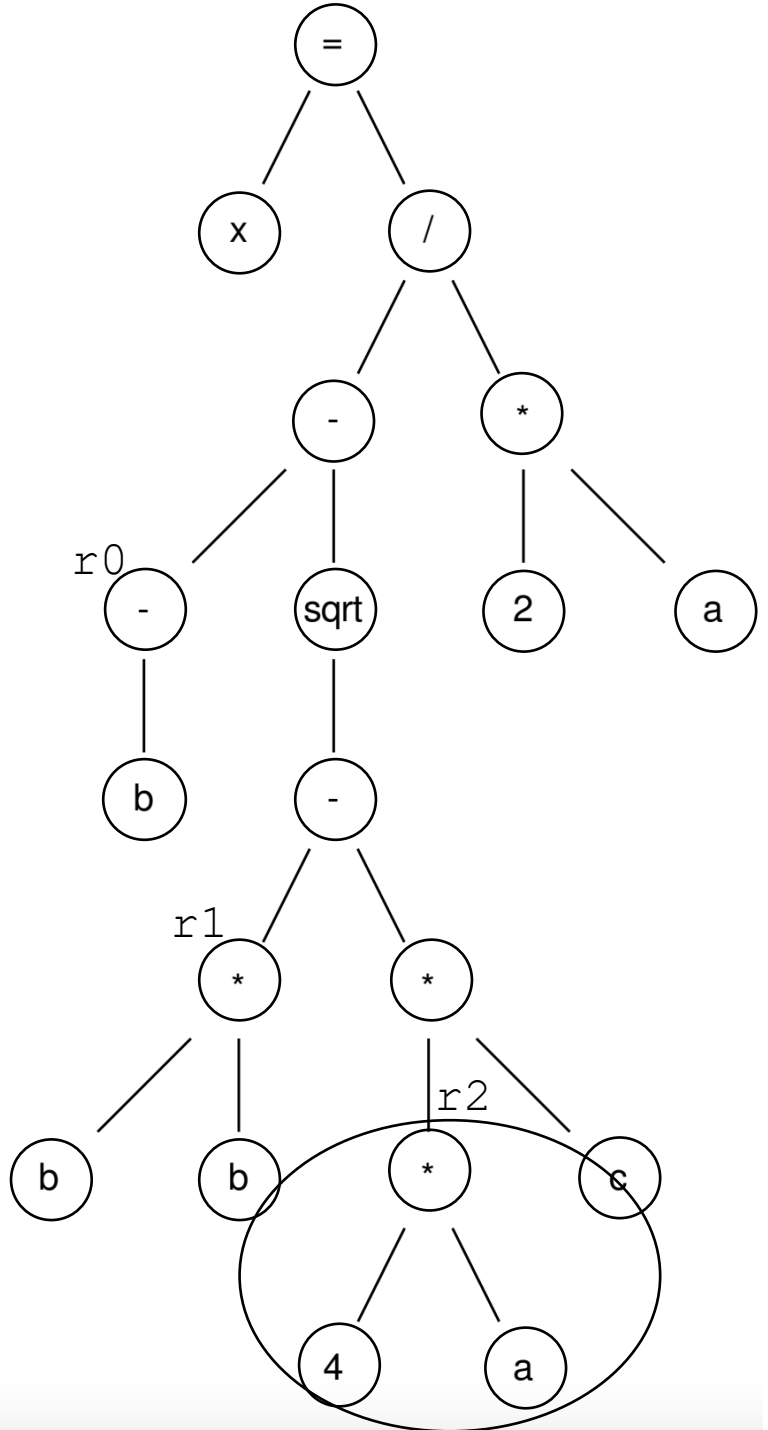
```
r0 = neg(b);
r1 = b * b;
```



Simplify this code:

post-order traversal, using temporary variables

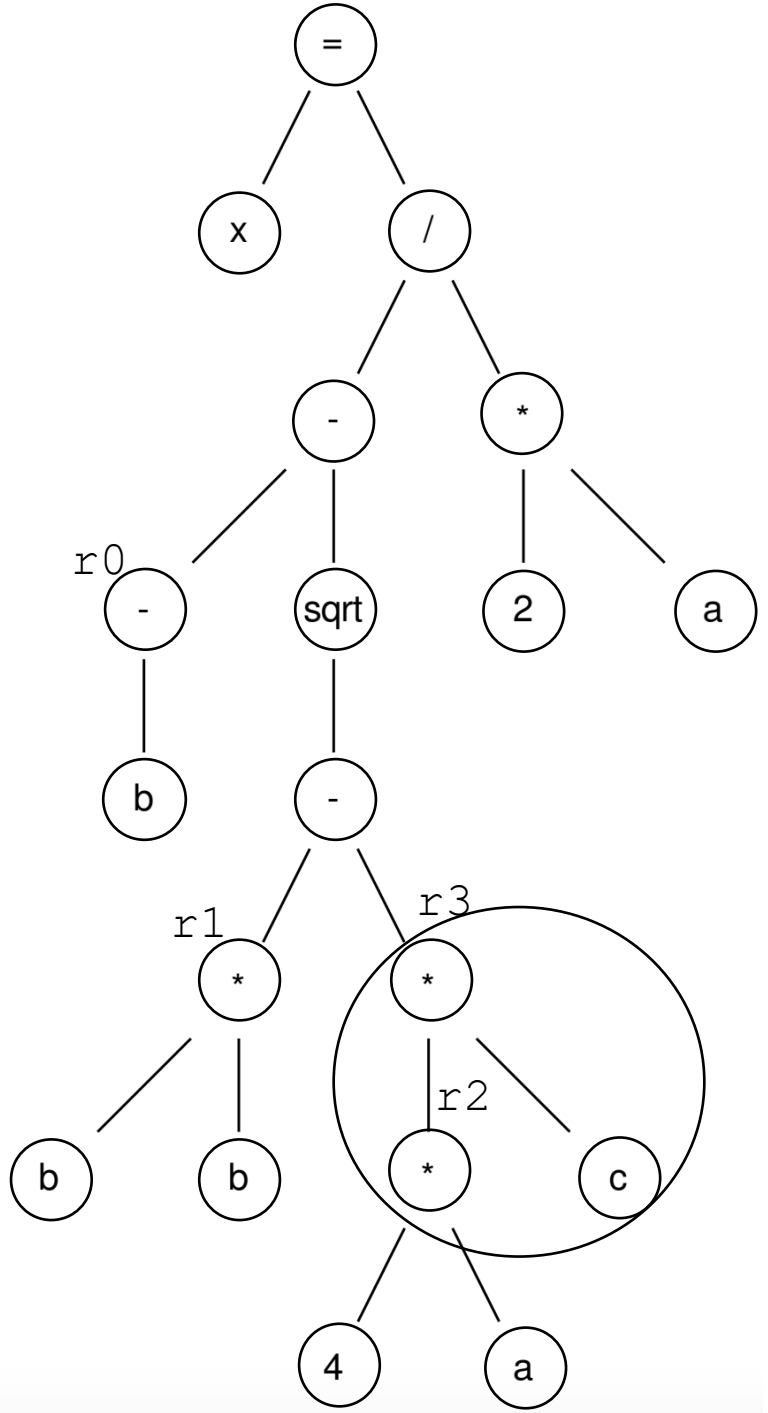
```
r0 = neg(b);
r1 = b * b;
r2 = 4 * a;
```



Simplify this code:

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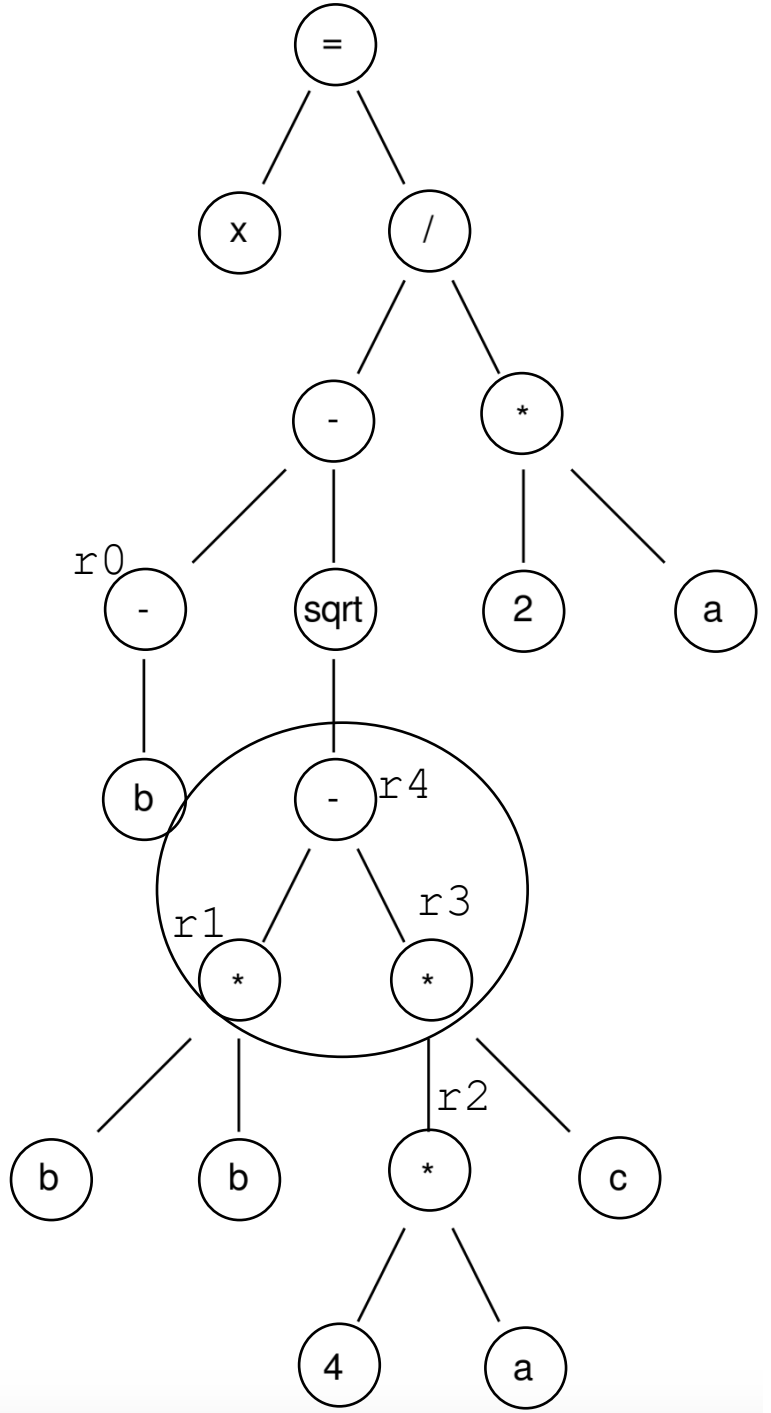
```
r0 = neg(b);
r1 = b * b;
r2 = 4 * a;
r3 = r2 * c;
```



Simplify this code:

post-order traversal, using temporary variables

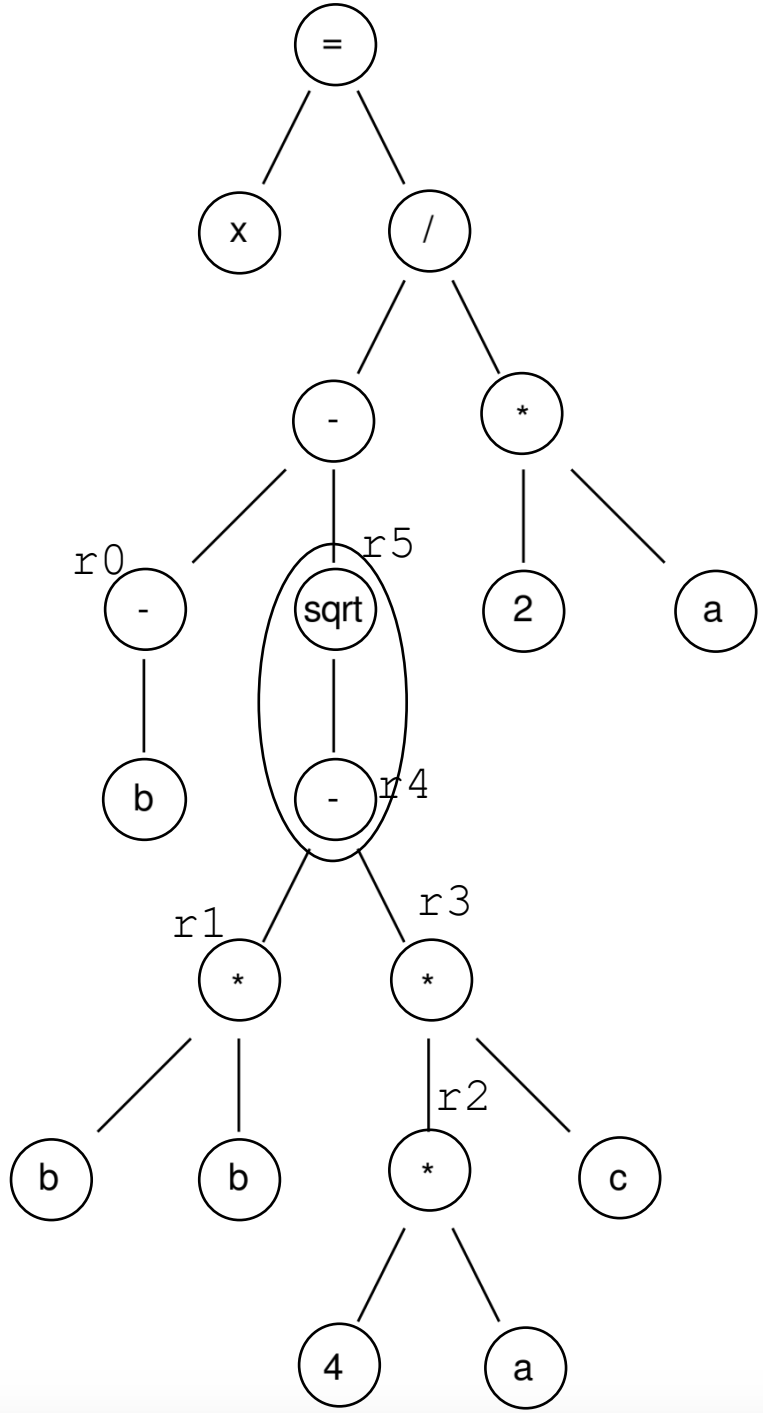
```
r0 = neg(b);
r1 = b * b;
r2 = 4 * a;
r3 = r2 * c;
r4 = r1 - r3;
```



Simplify this code:

post-order traversal, using temporary variables

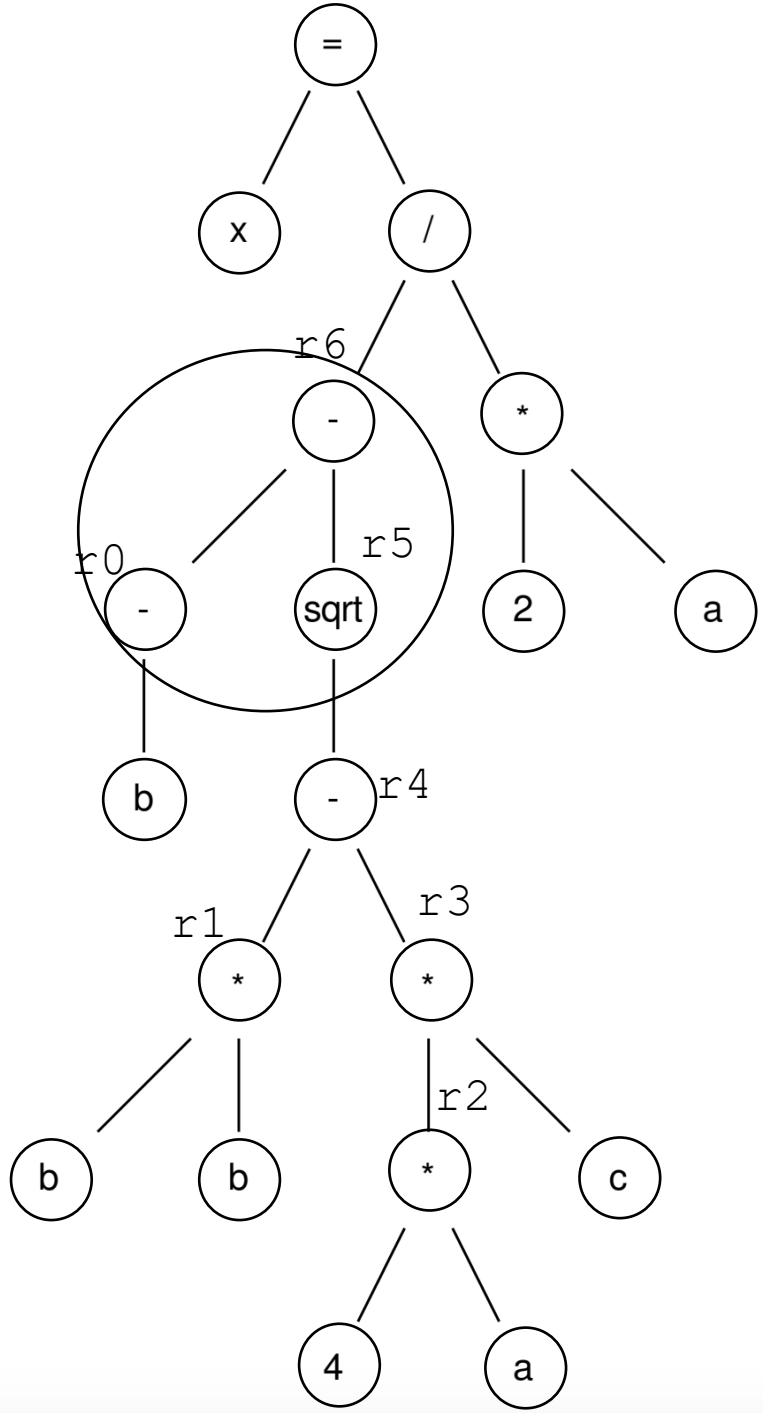
```
r0 = neg(b);
r1 = b * b;
r2 = 4 * a;
r3 = r2 * c;
r4 = r1 - r3;
r5 = sqrt(r4);
```



Simplify this code:

post-order traversal, using temporary variables

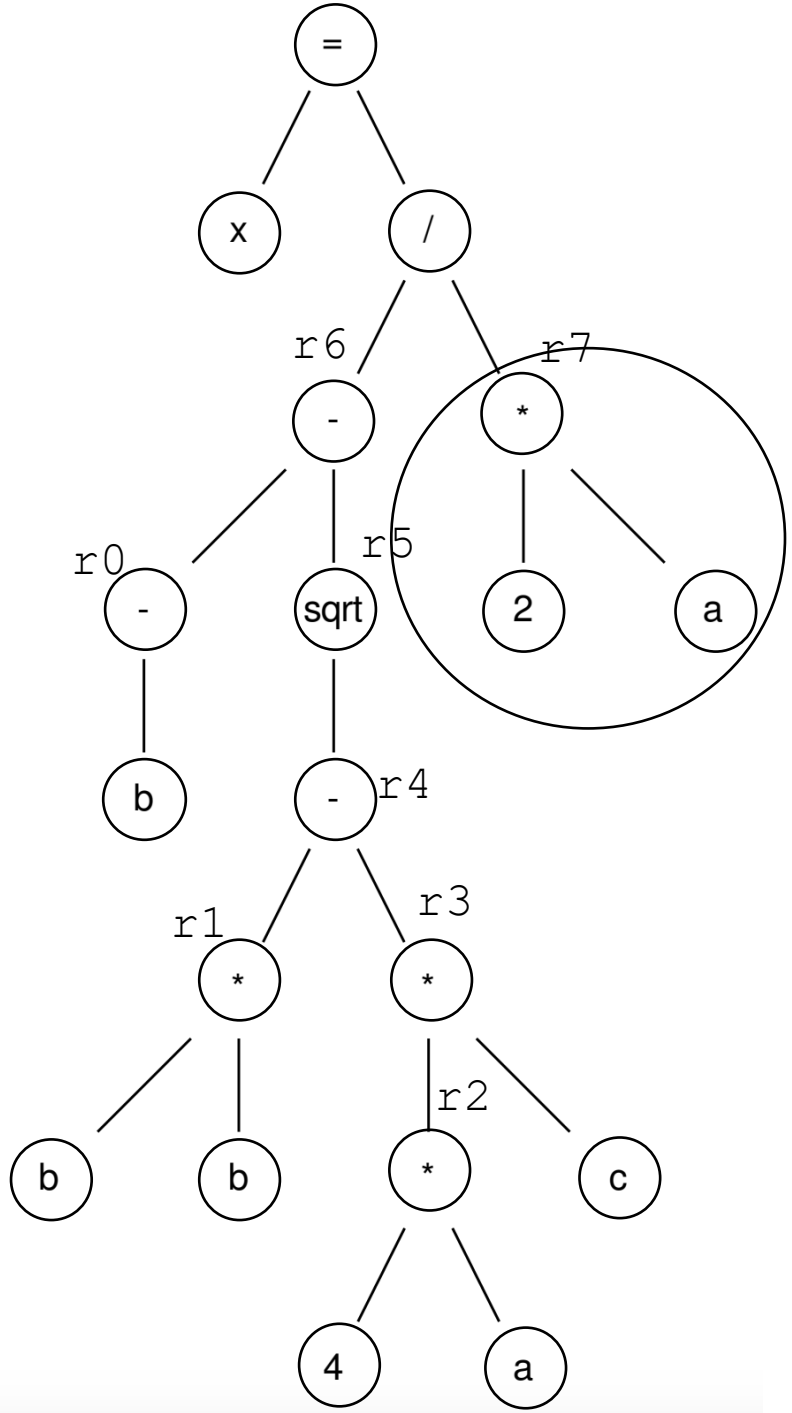
```
r0 = neg(b);
r1 = b * b;
r2 = 4 * a;
r3 = r2 * c;
r4 = r1 - r3;
r5 = sqrt(r4);
r6 = r0 - r5;
```



Simplify this code:

post-order traversal, using temporary variables

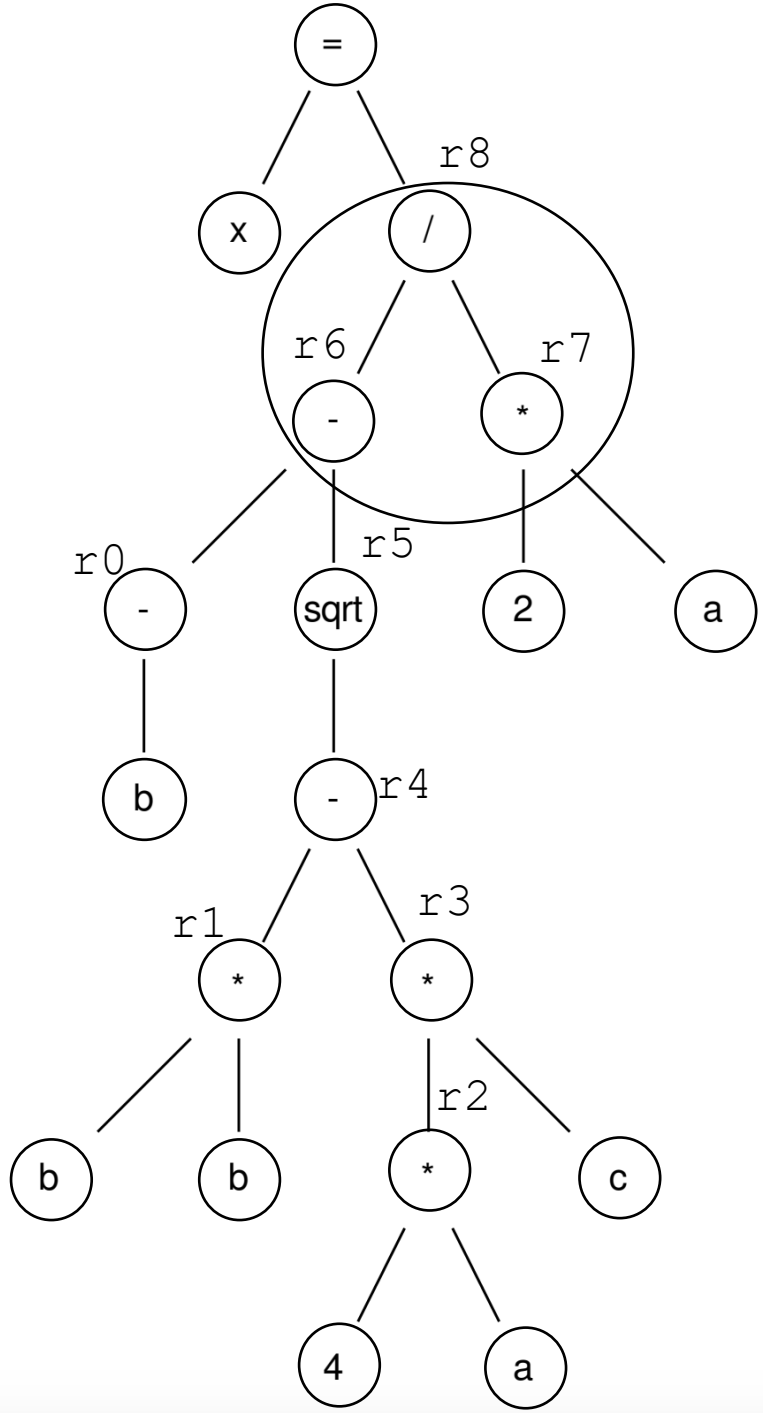
```
r0 = neg(b);
r1 = b * b;
r2 = 4 * a;
r3 = r2 * c;
r4 = r1 - r3;
r5 = sqrt(r4);
r6 = r0 - r5;
r7 = 2 * a;
```



Simplify this code:

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r0 = neg(b);
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r3 = r2 * c;
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r5 = sqrt(r4);
r6 = r0 - r5;
r7 = 2 * a;
r8 = r6 / r7;
```



Simplify this code:

post-order traversal, using temporary variables

```
r0 = neg(b);
r1 = b * b;
r2 = 4 * a;
r3 = r2 * c;
r4 = r1 - r3;
r5 = sqrt(r4);
r6 = r0 - r5;
r7 = 2 * a;
r8 = r6 / r7;
x  = r8;
```

- This is not exactly an ISA
  - unlimited registers
  - not always a 1-1 mapping of instructions.
- but it is much easier to translate to the ISA
- We call this an intermediate representation, or IR
- Examples of IR:
  LLVM, SPIR-V (IR for GPUs)

```c
// Type your code here, or load an example.
float sqrt(float x);

float add(float a, float b, float c) {
   return (-b - sqrt(b*b - 4 * a * c)) / (2*a);

}

```

A Output... Filter... Libraries + Add new... Add tool...

```
define dso_local float @_Z3addfff(float %0, float %1, float %2) #0 !dbg !
  %4 = alloca float, align 4
  %5 = alloca float, align 4
  %6 = alloca float, align 4
  store float %0, float* %4, align 4
  call void @llvm.dbg.declare(metadata float* %4, metadata !12, metadata
  store float %1, float* %5, align 4
  call void @llvm.dbg.declare(metadata float* %5, metadata !14, metadata
  store float %2, float* %6, align 4
  call void @llvm.dbg.declare(metadata float* %6, metadata !16, metadata
  %7 = load float, float* %5, align 4, !dbg !18
  %8 = fneg float %7, !dbg !19
  %9 = load float, float* %5, align 4, !dbg !20
  %10 = load float, float* %5, align 4, !dbg !21
  %11 = fmul float %9, %10, !dbg !22
  %12 = load float, float* %4, align 4, !dbg !23
  %13 = fmul float 4.000000e+00, %12, !dbg !24
  %14 = load float, float* %6, align 4, !dbg !25
  %15 = fmul float %13, %14, !dbg !26
  %16 = fsub float %11, %15, !dbg !27
  %17 = call float @_Z4sqrtf(float %16), !dbg !28
  %18 = fsub float %8, %17, !dbg !29
  %19 = load float, float* %4, align 4, !dbg !30
  %20 = fmul float 2.000000e+00, %19, !dbg !31
  %21 = fdiv float %18, %20, !dbg !32
  ret float %21, !dbg !33
}
```

C program

llvm IR

Memory accesses

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

Unless explicitly expressed in the programming language, loads and stores are split into multiple instructions!

Zoom out

- This can be a lot if you don’t have a compiler background; don’t feel overwhelmed!
- To be successful in this class, you don’t need to be an expert on compilation, ISAs, or IRs.
- The important thing is to have a mental model of how your complex code is broken down into instructions that are executed on hardware, especially loads and stores

New material – Instruction Level Parallelism

Instruction-level Parallelism (ILP)

- Parallelism from a **single stream of instructions**.
  - Output of program must match exactly a sequential execution!

- Widely applicable:
  - most mainstream programming languages are **sequential**
  - most deployed **hardware** has components to execute ILP

- Done by a combination of **programmer, compiler, and hardware**

Instruction-level Parallelism (ILP)

- What type of instructions can be executed in parallel?

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two instructions can be executed in parallel if they are independent

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two instructions can be executed in parallel if they are ***independent***

```
x = z + w;
a = b + c;
```

Two instructions are independent if the ***operand registers*** *are disjoint from the* ***result registers***

(assume all letter variables are registers)

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```
x = z + w;
a = b + x;
```

*Many times, dependencies can be easily tracked in the **compiler**:*

How can hardware execute ILP?

- Executing multiple instructions at once:

- Very Long Instruction Word (VLIW) architecture
  - Complicated architecture
  - Compiler combines multiple instructions into one

- Superscalar architecture:
  - Several sequential operations are issued in parallel

- Pipeline architecture:
  - Send instructions one after the other through stages

How can hardware execute ILP?

- Superscalar architecture:
  - Several sequential operations are issued in parallel
  - hardware detects dependencies

issue-width *is maximum number of instructions that can be issued in parallel*

```
instr0;
instr1;
instr2;
```

How can hardware execute ILP?

- Superscalar architecture:
  - Several sequential operations are issued in parallel
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issue-width *is maximum number of instructions that can be issued in parallel*

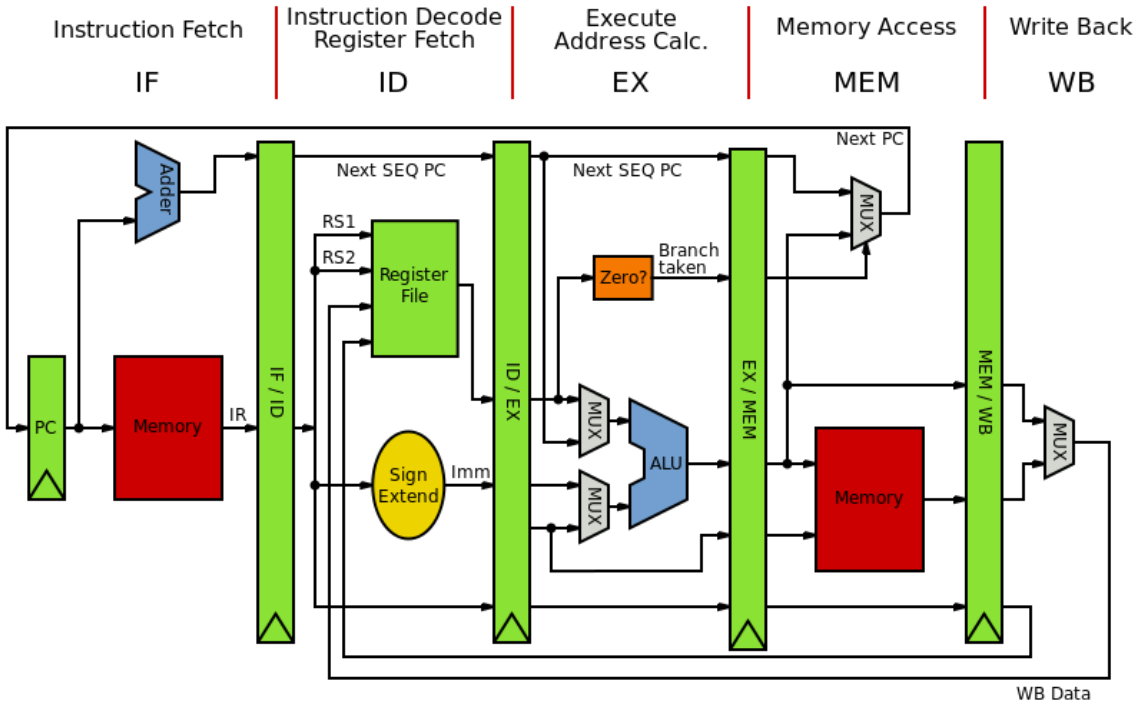
```
instr0;
instr1;
instr2;
```

if instr0 and instr1 are independent, they will be issued in parallel

How can hardware execute ILP?

- Pipeline parallelism

- Abstract mental model:
  - N-stage **pipeline**
  - N instructions can be in-flight
  - **Dependencies stall** pipeline

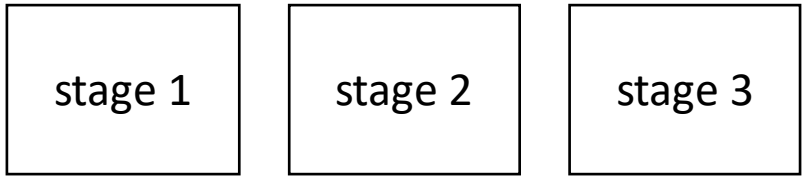


MIPS pipeline image from:
https://commons.wikimedia.org/wiki/Pipeline_(computer_hardware)

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
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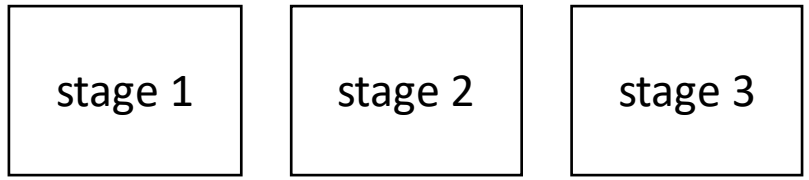



```
instr1;
instr2;
instr3;
```

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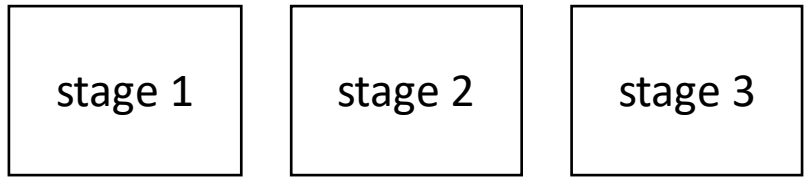



```
instr1;
instr2;
instr3;
```

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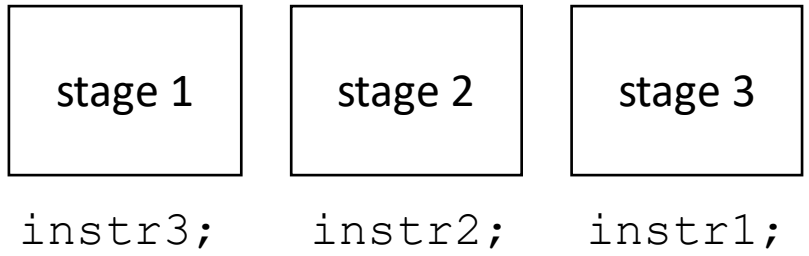

```
instr2;    instr1;
```

```
instr3;
```

Pipeline

- Pipeline parallelism

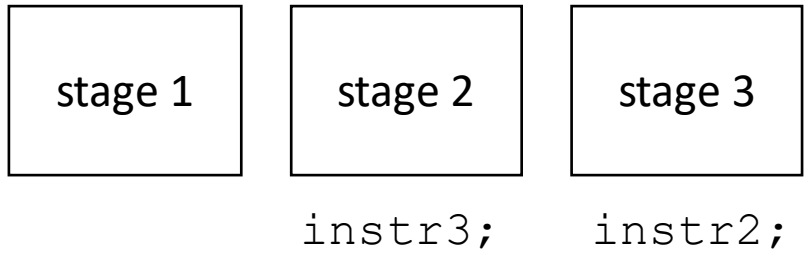
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Pipeline

- Pipeline parallelism

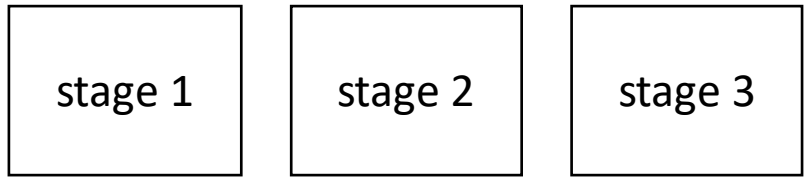
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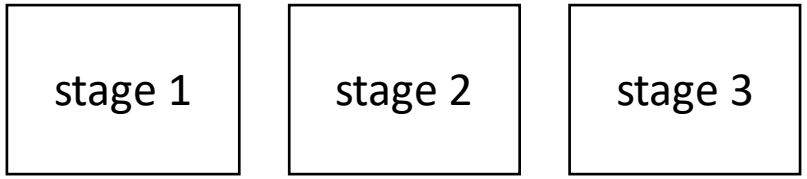



```
instr3;
```

Pipeline

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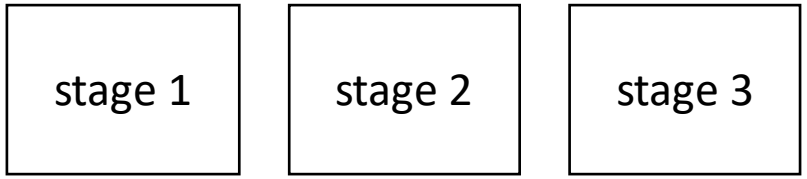
6 cycles for 3 independent instructions

Converges 1 instruction per cycle

Pipeline

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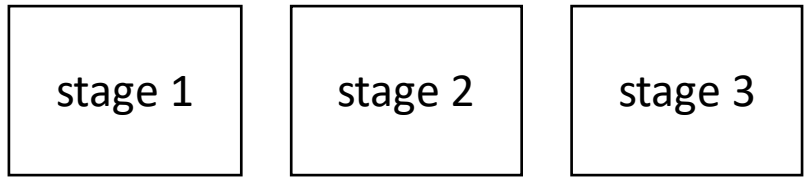

```
instr1;
instr2;
instr3;
```

What if the instructions depend on each other?

Pipeline

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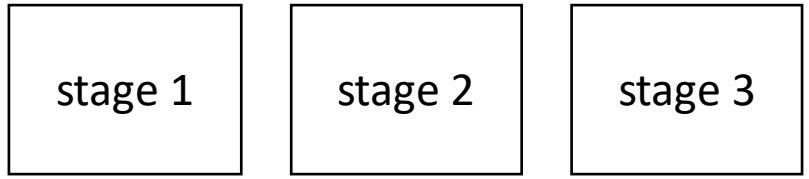

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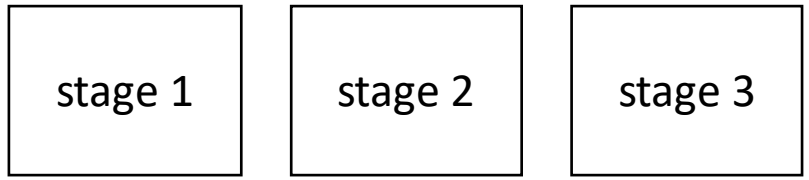

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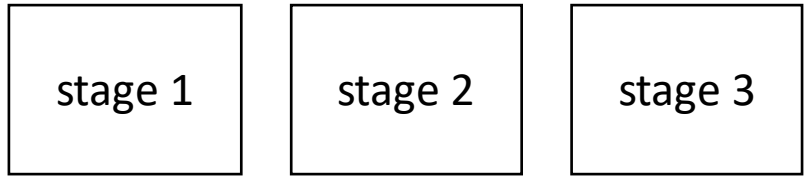

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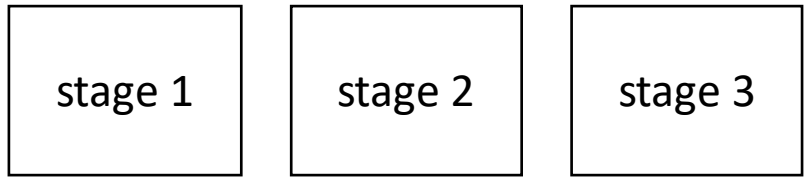



```
instr2;
instr3;
```

What if the instructions depend on each other?

Pipeline

- Pipeline parallelism
- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
  - Dependencies stall pipeline



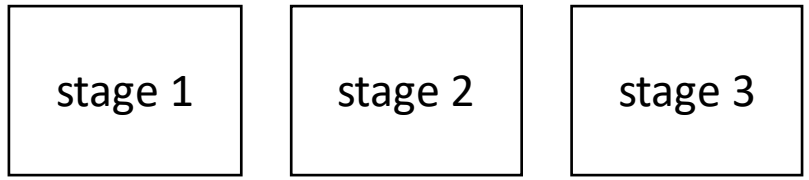

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```

```
instr3;
```

What if the instructions depend on each other?

Pipeline

- Pipeline parallelism
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```
instr2;
```

```
instr3;
```

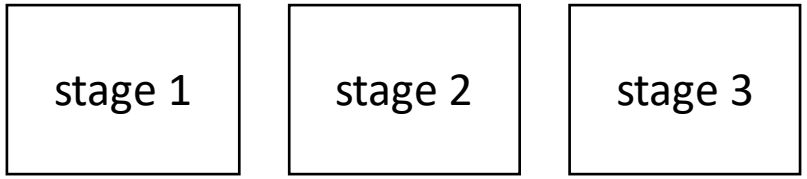
and so on...

What if the instructions depend on each other?

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
  - Dependencies stall pipeline



What if the instructions depend on each other?

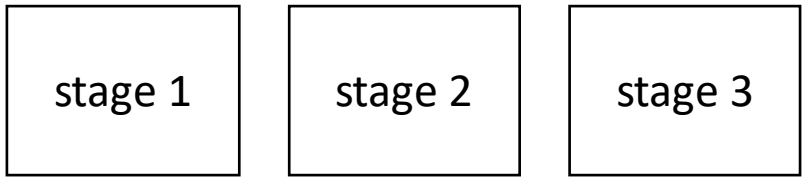
9 cycles for 3 instructions

converges to 3 cycles per instruction

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
  - Dependencies stall pipeline



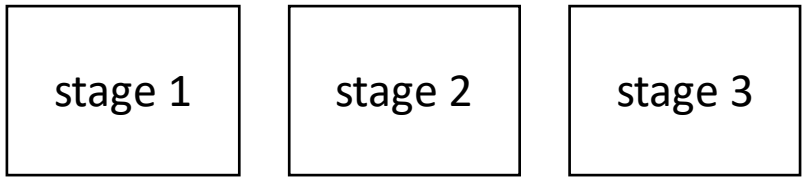

```
instr1;
instrX0;
instrX1;
instr2;
instrX2;
instrX3;
instr3;
```

If there are non-dependent instructions from other places in the program that we can interleave then we can get back performance!

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
  - Dependencies stall pipeline




```
instr1;
```

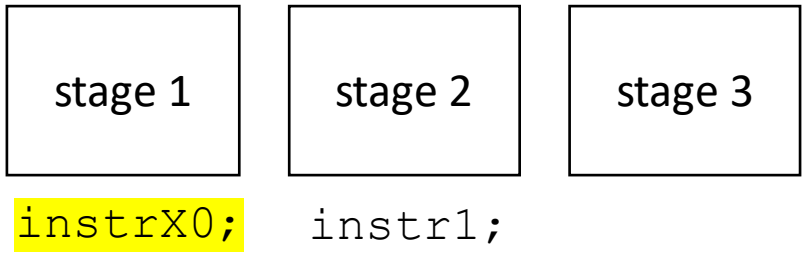
```
instrX0;
instrX1;
instr2;
instrX2;
instrX3;
instr3;
```

If there are non-dependent instructions from other places in the program that we can interleave then we can get back performance!

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
  - Dependencies stall pipeline




```
instrX1;
instr2;
instrX2;
instrX3;
instr3;
```

If there are non-dependent instructions from other places in the program that we can interleave then we can get back performance!

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
  - Dependencies stall pipeline




```
instr2;
instrX2;
instrX3;
instr3;
```

If there are non-dependent instructions from other places in the program that we can interleave then we can get back performance!

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
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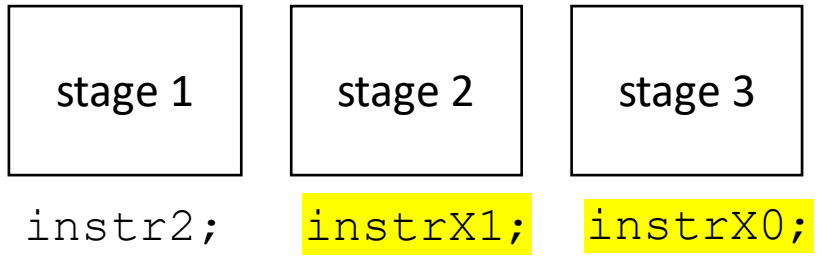

```
instrX2;
instrX3;
instr3;
```

If there are non-dependent instructions from other places in the program that we can interleave then we can get back performance!

Pipeline

- Pipeline parallelism

- Abstract mental model for compiler:
  - N-stage pipeline
  - N instructions can be in-flight
  - Dependencies stall pipeline




```
instrX2;
instrX3;
instr3;
```

and so on...

We converge to 1 cycle per instruction again!

If there are non-dependent instructions from other places in the program that we can interleave then we can get back performance!

Independent Instructions

- Out-of-order execution
  - Hardware looks ahead for independent instructions
  - Hardware delays dependent instructions

What does this look like in the real world?

- Intel Haswell (2013):
  - Issue width of 4
  - 14-19 stage pipeline
  - OoO execution
- Intel Nehalem (2008)
  - Issue width of 2-4
  - 20-24 stage pipeline
  - OoO execution
- ARM
  - Cortex V8 has issue width of 2
  - V7 has 3 stage pipeline; Cortex V8 has 13
  - OoO execution
- RISC-V
  - Ariane and Rocket (open-source processor implementations) are In-Order
  - 3-6 stage pipelines
  - some superscalar implementations (BOOM)

What does this mean for us?

- We should have an abstract performance model for instruction scheduling (the order of instructions)
- Try not to place dependent instructions in sequence
- The compiler will often help us here, but sometimes it cannot!

Two techniques to optimize for ILP

- Independent for loops (loop unrolling)
- Reduction for loops (loop unrolling)

What is loop unrolling?

```
for (int i = 0;  i < 12;  i++) {
  a[i] = b[i] + c[i];
}
```

Can we unroll this loop?
Data and **control** dependencies

```
for (int i = 0;  i < 12;  i+=2) {
  a[i] = b[i] + c[i];
  a[i+1] = b[i+1] + c[i+1];
}
```

Using Loop Unrolling to Exploit ILP

- for loops with independent chains of computation

```
for (int i = 0; i < SIZE; i++) {
    SEQ(i);
}
```

where:

```
SEQ(i) = instr1;
         instr2;
         ...
         a[i] = instrN;
```

and let instr(N) depend on instr(N-1)

loops only write to memory addressed by the loop variable

Using Loop Unrolling to Exploit ILP

- Simple loop unrolling:

```
for (int i = 0; i < SIZE; i+=2) {
    SEQ(i);
    SEQ(i+1);
}
```

Saves one addition and one comparison per loop, but doesn't help with ILP

Using Loop Unrolling to Exploit ILP

- Simple loop unrolling:

```
for (int i = 0; i < SIZE; i+=2) {
    SEQ(i);
    SEQ(i+1);
}
```

Let green highlights indicate instructions from iteration `i`.

Let blue highlights indicate instructions from iteration `i + 1`.

Using Loop Unrolling to Exploit ILP

- Simple loop unrolling:

```
for (int i = 0; i < SIZE; i+=2) {
    SEQ(i);
    SEQ(i+1);
}
```

Let `SEQ(i,j)` be the jth instruction of `SEQ(i)`.

Let each instruction chain have N instructions

Using Loop Unrolling to Exploit ILP

- Simple loop unrolling:

```
for (int i = 0; i < SIZE; i+=2) {
     SEQ(i,1);
     SEQ(i,2);
     ...
     SEQ(i,N); // end iteration for i
     SEQ(i+1,1);
     SEQ(i+1,2);
     ...
     SEQ(i+1, N); // end iteration for i + 1
}
```

Let `SEQ(i,j)` be the jth instruction of `SEQ(i)`.

Let each instruction chain have N instructions

Using Loop Unrolling to Exploit ILP

- Simple loop unrolling:

```
for (int i = 0; i < SIZE; i+=2) {
    SEQ(i,1);
    SEQ(i+1,1);
    SEQ(i,2);
    SEQ(i+1,2);
    ...
    SEQ(i,N);
    SEQ(i+1, N);
}
```

They can be interleaved

Using Loop Unrolling to Exploit ILP

- Simple loop unrolling:

```
for (int i = 0; i < SIZE; i+=2) {
    SEQ(i,1);
    SEQ(i+1,1);
    SEQ(i,2);
    SEQ(i+1,2);
    ...
    SEQ(i,N);
    SEQ(i+1, N);
}
```

They can be interleaved

two instructions can be pipelined, or executed on a superscalar processor

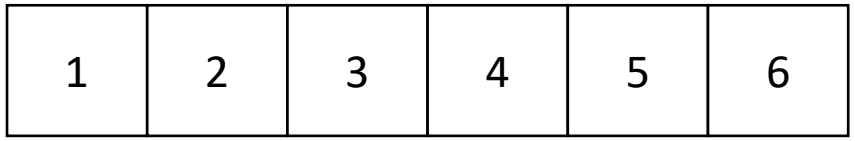
Using Loop Unrolling to Exploit ILP

- This is what you are doing in part 1 of homework 1
- You are playing the role of a compiler unrolling loops
- Your “compiler” is written in Python. You print out C++ code
- Your code is parameterized by dependency chain and by unroll factor

Loop Unrolling for Reduction Loops

- Prior approach examined loops with independent iterations and chains of dependent computations

- Now we will look at reduction loops:
  - Entire computation is dependent
  - Typically, short bodies (addition, multiplication, max, min)



addition: 21

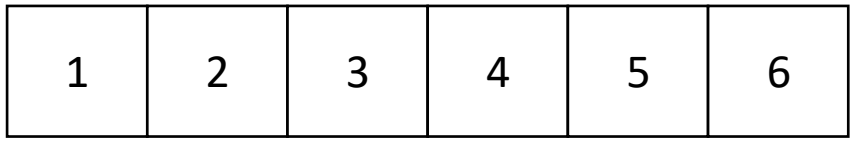
max: 6

min: 1

Loop Unrolling for Reduction Loops

- Simple implementation:

```
for (int i = 1; i < SIZE; i++) {
    a[0] = REDUCE(a[0], a[i]);
}
```



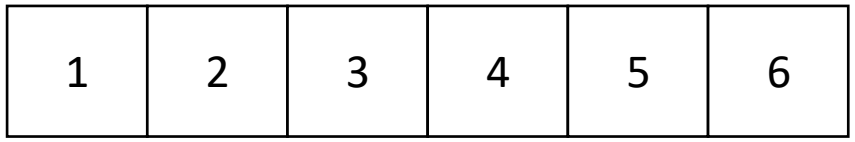
1 + 2 + 3 + 4 + 5 + 6

Loop Unrolling for Reduction Loops

- Simple implementation:

```
for (int i = 1; i < SIZE; i++) {
    a[0] = REDUCE(a[0], a[i]);
}
```

What is associativity?



(1 + 2 + 3) + (4 + 5 + 6)

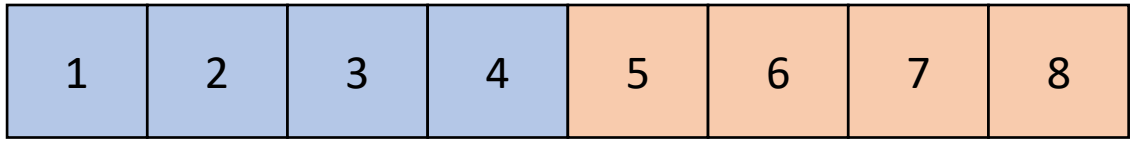
Loop Unrolling for Reduction Loops

- Chunk array in equal sized partitions and do local reductions
- Consider size 2:

1	2	3	4	5	6	7	8

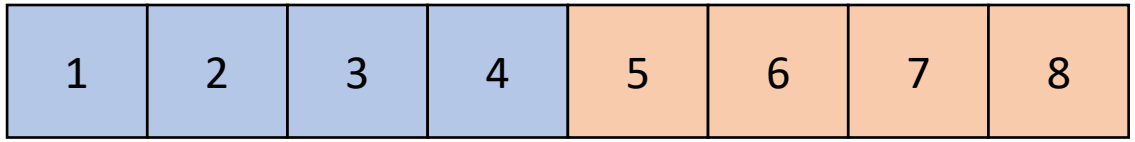
Loop Unrolling for Reduction Loops

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Loop Unrolling for Reduction Loops

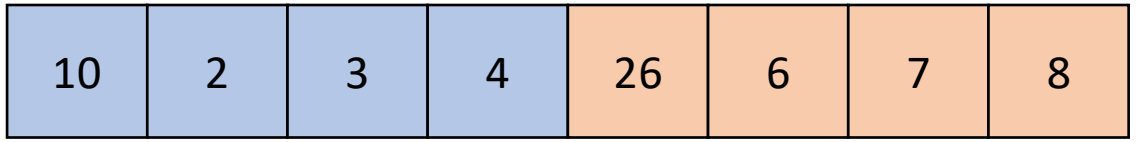
- chunk array in equal sized partitions and do local reductions
- Consider size 2:



Do addition reduction in base memory location

Loop Unrolling for Reduction Loops

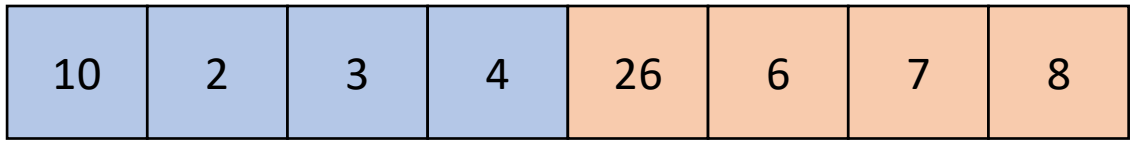
- chunk array in equal sized partitions and do local reductions
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Do addition reduction in base memory location

Loop Unrolling for Reduction Loops

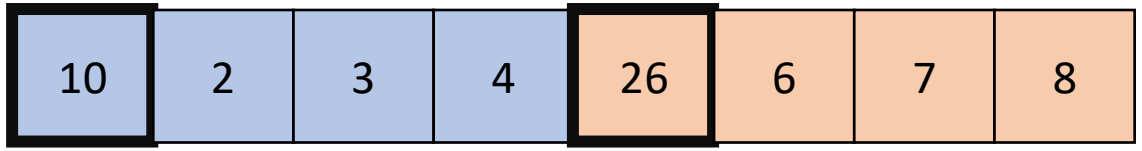
- chunk array in equal sized partitions and do local reductions
- Consider size 2:



Add together base locations

Loop Unrolling for Reduction Loops

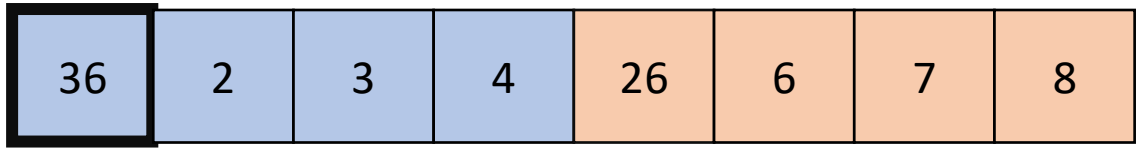
- chunk array in equal sized partitions and do local reductions
- Consider size 2:



Add together base locations

Loop Unrolling for Reduction Loops

- chunk array in equal sized partitions and do local reductions
- Consider size 2:



Add together base locations

Loop Unrolling for Reduction Loops

- Simple implementation:

```
for (int i = 1; i < SIZE/2; i++) {
    a[0] = REDUCE(a[0], a[i]);
    a[SIZE/2] = REDUCE(a[SIZE/2], a[(SIZE/2)+i]);
}

a[0] = REDUCE(a[0], a[SIZE/2])
```

Loop Unrolling for Reduction Loops

- Simple implementation:

```
for (int i = 1; i < SIZE/2; i++) {
    a[0] = REDUCE(a[0], a[i]);
    a[SIZE/2] = REDUCE(a[SIZE/2], a[(SIZE/2)+i]);
}

a[0] = REDUCE(a[0], a[SIZE/2])
```

Loop Unrolling for Reduction Loops

- Simple implementation:

```
for (int i = 1; i < SIZE/2; i++) {
    a[0] = REDUCE(a[0], a[i]);
    a[SIZE/2] = REDUCE(a[SIZE/2], a[(SIZE/2)+i]);
}

a[0] = REDUCE(a[0], a[SIZE/2])
```

independent instructions can be done in parallel!

Watch out!

- Our abstraction: separate dependent instructions as far as possible

- Pros:
  - Simple

- Cons:
  - Can lead to register spilling, causing expensive loads

consider `instr1` and `instr2` have a data dependence, and `instrX's` are independent

```
instr1;
instrX0;
instrX1;
...
Instr2;
```

independent instructions. If they overwrite the register storing `instr1's` result, then it will have to be stored to memory and retrieved before `instr2`

Watch out!

- Our abstraction: separate dependent instructions as far as possible

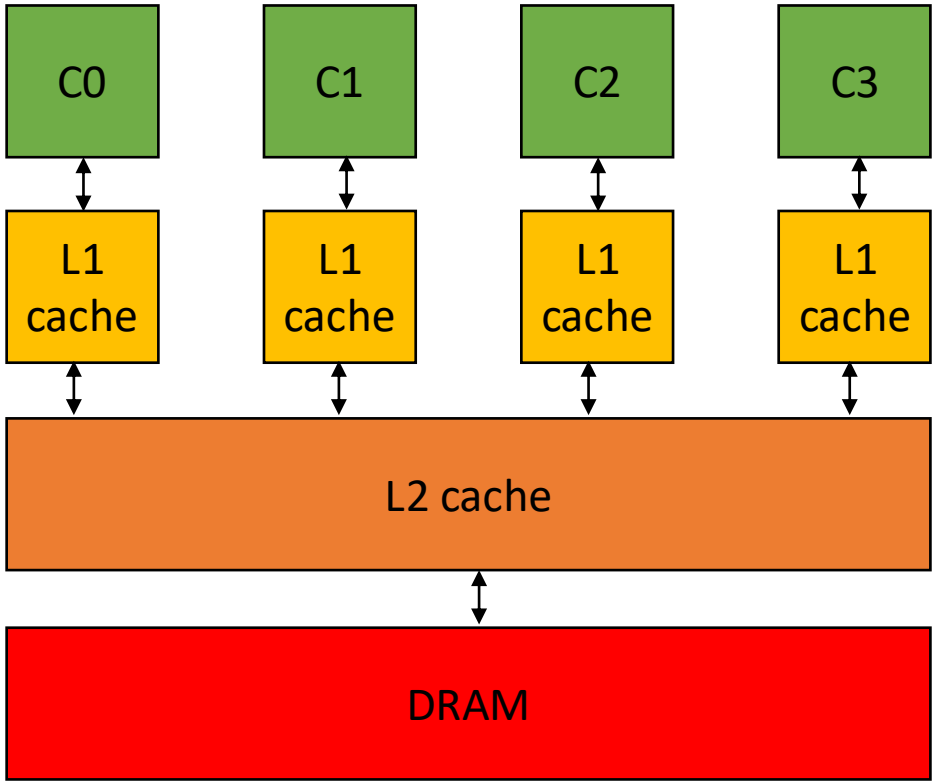
- Pros:
  - Simple

- Cons:
  - Can lead to register spilling, causing expensive loads

Solutions include using a **resource model** to guide the topological ordering. Highly architecture dependent. Compiler algorithms become more expensive.

Consider timing the compile time in your homework assignment.

Memory hierarchy overview


C0
C1
C2
C3
L1 cache
L1 cache
L1 cache
L1 cache
L2 cache
DRAM

Core

A core executes a stream
of sequential ISA instructions

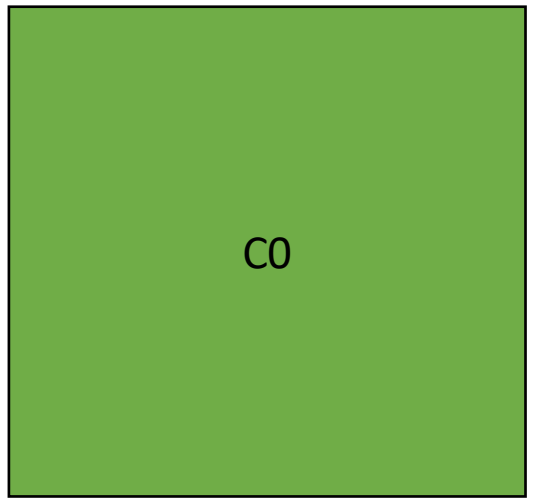
A good mental model executes
1 ISA instruction per cycle

3 Ghz means 3B cycles per second
1 ISA instruction takes .33 ns

Compiled function #0

```
13        movd    eax, xmm0
14        xor     eax, 2147483648
15        movd    xmm0, eax
16        movss   dword ptr [rbp - 16], xmm0
17        movss   xmm0, dword ptr [rbp - 8]
18        mulss   xmm0, dword ptr [rbp - 8]
19        movss   xmm1, dword ptr [rip + .LCPI0_1]
20        mulss   xmm1, dword ptr [rbp - 4]
21        mulss   xmm1, dword ptr [rbp - 12]
22        subss   xmm0, xmm1
23        call    sqrt(float)
24        movaps  xmm1, xmm0
25        movss   xmm0, dword ptr [rbp - 16]
26        subss   xmm0, xmm1
27        movss   xmm1, dword ptr [rip + .LCPI0_0]
28        mulss   xmm1, dword ptr [rbp - 4]
29        divss   xmm0, xmm1
```

Thread 0



Core

Core

Sometimes multiple programs want to share the same core.

Compiled function #0

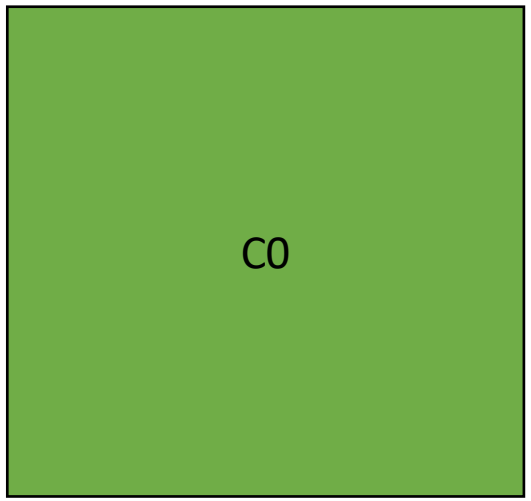
```
13      movd    eax, xmm0
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15      movd    xmm0, eax
16      movss   dword ptr [rbp - 16], xmm0
17      movss   xmm0, dword ptr [rbp - 8]
18      mulss   xmm0, dword ptr [rbp - 8]
19      movss   xmm1, dword ptr [rip + .LCPI0_1]
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22      subss   xmm0, xmm1
23      call    sqrt(float)
24      movaps  xmm1, xmm0
25      movss   xmm0, dword ptr [rbp - 16]
26      subss   xmm0, xmm1
27      movss   xmm1, dword ptr [rip + .LCPI0_0]
28      mulss   xmm1, dword ptr [rbp - 4]
29      divss   xmm0, xmm1
```

Thread 0

Compiled function #1

```
        movss   xmm0, dword ptr [rbp - 8]
        mulss   xmm0, dword ptr [rbp - 8]
        movss   xmm1, dword ptr [rip + .LCPI0_1]
        mulss   xmm1, dword ptr [rbp - 4]
        mulss   xmm1, dword ptr [rbp - 12]
        subss   xmm0, xmm1
        call    sqrt(float)
        movaps  xmm1, xmm0
        movss   xmm0, dword ptr [rbp - 16]
        subss   xmm0, xmm1
        movss   xmm1, dword ptr [rip + .LCPI0_0]
        mulss   xmm1, dword ptr [rbp - 4]
        divss   xmm0, xmm1
        add     rsp, 16
```

Thread 1



Core

Core

Sometimes multiple programs want to share the same core.

Compiled function #0

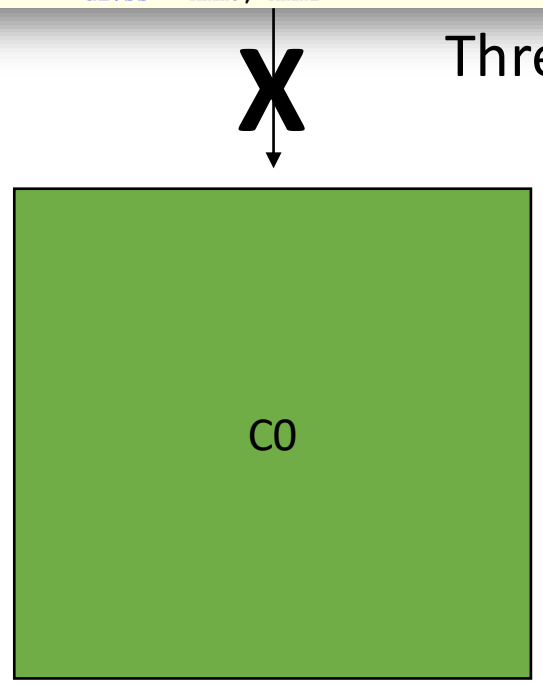
```
13        movd    eax, xmm0
14        xor     eax, 2147483648
15        movd    xmm0, eax
16        movss   dword ptr [rbp - 16], xmm0
17        movss   xmm0, dword ptr [rbp - 8]
18        mulss   xmm0, dword ptr [rbp - 8]
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20        mulss   xmm1, dword ptr [rbp - 4]
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26        subss   xmm0, xmm1
27        movss   xmm1, dword ptr [rip + .LCPI0_0]
28        mulss   xmm1, dword ptr [rbp - 4]
29        divss   xmm0, xmm1
```

Thread 0

Compiled function #1

```
        movss   xmm0, dword ptr [rbp - 8]
        mulss   xmm0, dword ptr [rbp - 8]
        movss   xmm1, dword ptr [rip + .LCPI0_1]
        mulss   xmm1, dword ptr [rbp - 4]
        mulss   xmm1, dword ptr [rbp - 12]
        subss   xmm0, xmm1
        call    sqrt(float)
        movaps  xmm1, xmm0
        movss   xmm0, dword ptr [rbp - 16]
        subss   xmm0, xmm1
        movss   xmm1, dword ptr [rip + .LCPI0_0]
        mulss   xmm1, dword ptr [rbp - 4]
        divss   xmm0, xmm1
        add     rsp, 16
```

Thread 1



Core



The OS can preempt a thread
(remove it from the hardware resource)

Core

Compiled function #1

```
movss   xmm0, dword ptr [rbp - 8]       #
mulss   xmm0, dword ptr [rbp - 8]
movss   xmm1, dword ptr [rip + .LCPI0_1] #
mulss   xmm1, dword ptr [rbp - 4]
mulss   xmm1, dword ptr [rbp - 12]
subss   xmm0, xmm1
call    sqrt(float)
movaps  xmm1, xmm0
movss   xmm0, dword ptr [rbp - 16]      #
subss   xmm0, xmm1
movss   xmm1, dword ptr [rip + .LCPI0_0] #
mulss   xmm1, dword ptr [rbp - 4]
divss   xmm0, xmm1
add     rsp, 16
```

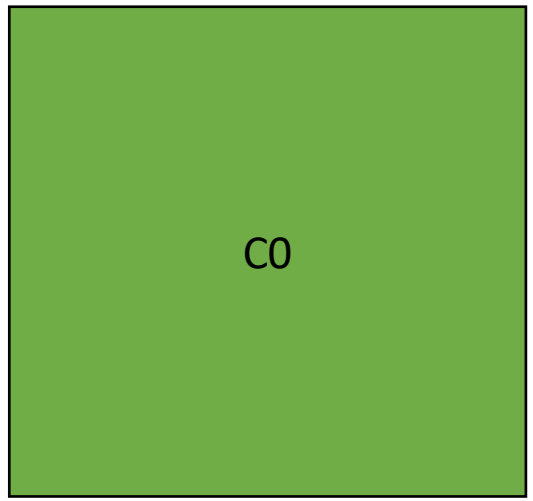
Thread 1

Compiled function #0

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13      movd    eax, xmm0
14      xor     eax, 2147483648
15      movd    xmm0, eax
16      movss   dword ptr [rbp - 16], xmm0
17      movss   xmm0, dword ptr [rbp - 8]
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19      movss   xmm1, dword ptr [rip + .LCPI0_1]
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27      movss   xmm1, dword ptr [rip + .LCPI0_0]
28      mulss   xmm1, dword ptr [rbp - 4]
29      divss   xmm0, xmm1
```

Thread 0

Sometimes multiple programs want to share the same core.



Core



And place another thread to execute

This is called concurrency: multiple threads taking turns executing on the same hardware resource

Core

Compiled function #1

```
movss   xmm0, dword ptr [rbp - 8]
mulss   xmm0, dword ptr [rbp - 8]
movss   xmm1, dword ptr [rip + .LCPI0_1]
mulss   xmm1, dword ptr [rbp - 4]
mulss   xmm1, dword ptr [rbp - 12]
subss   xmm0, xmm1
call    sqrt(float)
movaps  xmm1, xmm0
movss   xmm0, dword ptr [rbp - 16]
subss   xmm0, xmm1
movss   xmm1, dword ptr [rip + .LCPI0_0]
mulss   xmm1, dword ptr [rbp - 4]
divss   xmm0, xmm1
add     rsp, 16
```

Thread 1

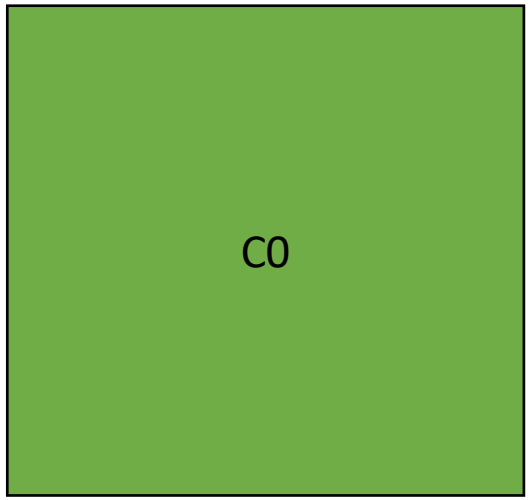
Compiled function #0

```
13      movd    eax, xmm0
14      xor     eax, 2147483648
15      movd    xmm0, eax
16      movss   dword ptr [rbp - 16], xmm0
17      movss   xmm0, dword ptr [rbp - 8]
18      mulss   xmm0, dword ptr [rbp - 8]
19      movss   xmm1, dword ptr [rip + .LCPI0_1]
20      mulss   xmm1, dword ptr [rbp - 4]
21      mulss   xmm1, dword ptr [rbp - 12]
22      subss   xmm0, xmm1
23      call    sqrt(float)
24      movaps  xmm1, xmm0
25      movss   xmm0, dword ptr [rbp - 16]
26      subss   xmm0, xmm1
27      movss   xmm1, dword ptr [rip + .LCPI0_0]
28      mulss   xmm1, dword ptr [rbp - 4]
29      divss   xmm0, xmm1
```

Thread 0

Preemption can occur:

- when a thread executes a long latency instruction
- periodically from the OS to provide fairness
- explicitly using sleep instructions



Core



And place another thread to execute

Multicores

Compiled function #0

```
13        movd    eax, xmm0
14        xor     eax, 2147483648
15        movd    xmm0, eax
16        movss   dword ptr [rbp - 16], xmm0
17        movss   xmm0, dword ptr [rbp - 8]
18        mulss   xmm0, dword ptr [rbp - 8]
19        movss   xmm1, dword ptr [rip + .LCPI0_1]
20        mulss   xmm1, dword ptr [rbp - 4]
21        mulss   xmm1, dword ptr [rbp - 12]
22        subss   xmm0, xmm1
23        call    sqrt(float)
24        movaps  xmm1, xmm0
25        movss   xmm0, dword ptr [rbp - 16]
26        subss   xmm0, xmm1
27        movss   xmm1, dword ptr [rip + .LCPI0_0]
28        mulss   xmm1, dword ptr [rbp - 4]
29        divss   xmm0, xmm1
```

Compiled function #1

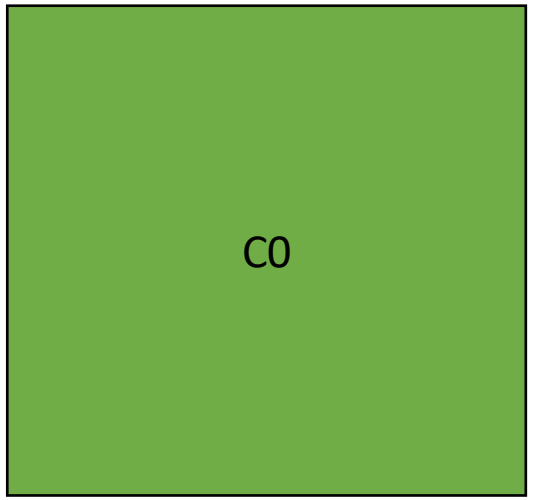
```
        movss   xmm0, dword ptr [rbp - 8]
        mulss   xmm0, dword ptr [rbp - 8]
        movss   xmm1, dword ptr [rip + .LCPI0_1]
        mulss   xmm1, dword ptr [rbp - 4]
        mulss   xmm1, dword ptr [rbp - 12]
        subss   xmm0, xmm1
        call    sqrt(float)
        movaps  xmm1, xmm0
        movss   xmm0, dword ptr [rbp - 16]
        subss   xmm0, xmm1
        movss   xmm1, dword ptr [rip + .LCPI0_0]
        mulss   xmm1, dword ptr [rbp - 4]
        divss   xmm0, xmm1
        add     rsp, 16
```

Thread 0

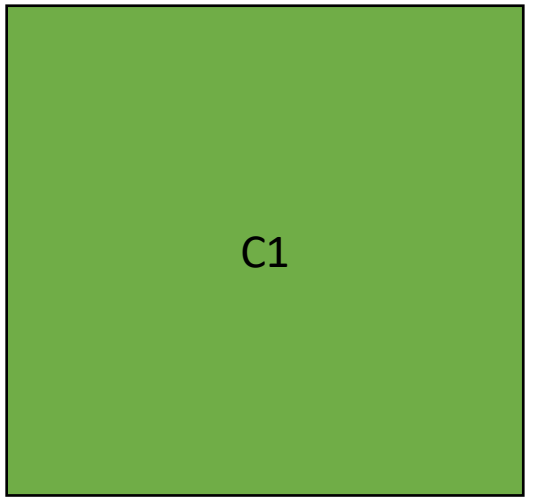
Thread 1

Threads can execute simultaneously (at the same time) if there are enough resources.

This is also concurrency. But when they execute at the same time, it's called: parallelism.



Core



Core

Multicores

This is fine if threads are independent: e.g. running Chrome and Spotify at the same time.

If threads need to cooperate to run the program, then they need to communicate through memory

Compiled function #0

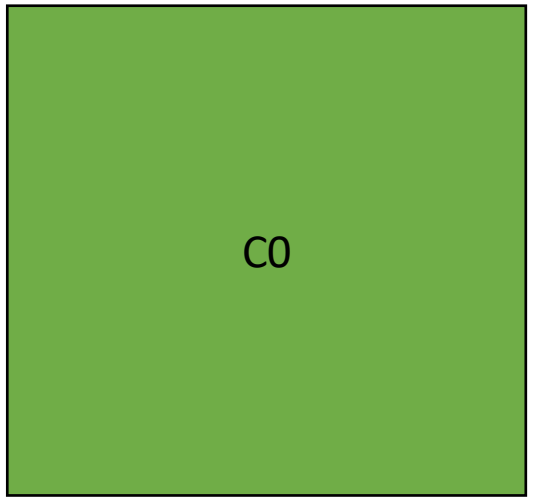
```
13      movd    eax, xmm0
14      xor     eax, 2147483648
15      movd    xmm0, eax
16      movss   dword ptr [rbp - 16], xmm0
17      movss   xmm0, dword ptr [rbp - 8]
18      mulss   xmm0, dword ptr [rbp - 8]
19      movss   xmm1, dword ptr [rip + .LCPI0_1]
20      mulss   xmm1, dword ptr [rbp - 4]
21      mulss   xmm1, dword ptr [rbp - 12]
22      subss   xmm0, xmm1
23      call    sqrt(float)
24      movaps  xmm1, xmm0
25      movss   xmm0, dword ptr [rbp - 16]
26      subss   xmm0, xmm1
27      movss   xmm1, dword ptr [rip + .LCPI0_0]
28      mulss   xmm1, dword ptr [rbp - 4]
29      divss   xmm0, xmm1
```

Thread 0

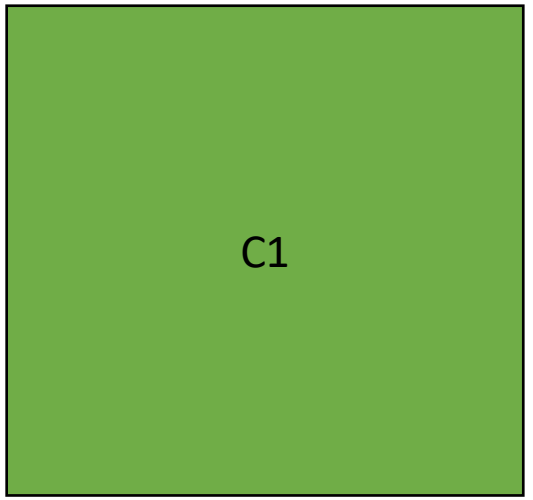
Compiled function #1

```
        movss   xmm0, dword ptr [rbp - 8]
        mulss   xmm0, dword ptr [rbp - 8]
        movss   xmm1, dword ptr [rip + .LCPI0_1]
        mulss   xmm1, dword ptr [rbp - 4]
        mulss   xmm1, dword ptr [rbp - 12]
        subss   xmm0, xmm1
        call    sqrt(float)
        movaps  xmm1, xmm0
        movss   xmm0, dword ptr [rbp - 16]
        subss   xmm0, xmm1
        movss   xmm1, dword ptr [rip + .LCPI0_0]
        mulss   xmm1, dword ptr [rbp - 4]
        divss   xmm0, xmm1
        add     rsp, 16
```

Thread 1



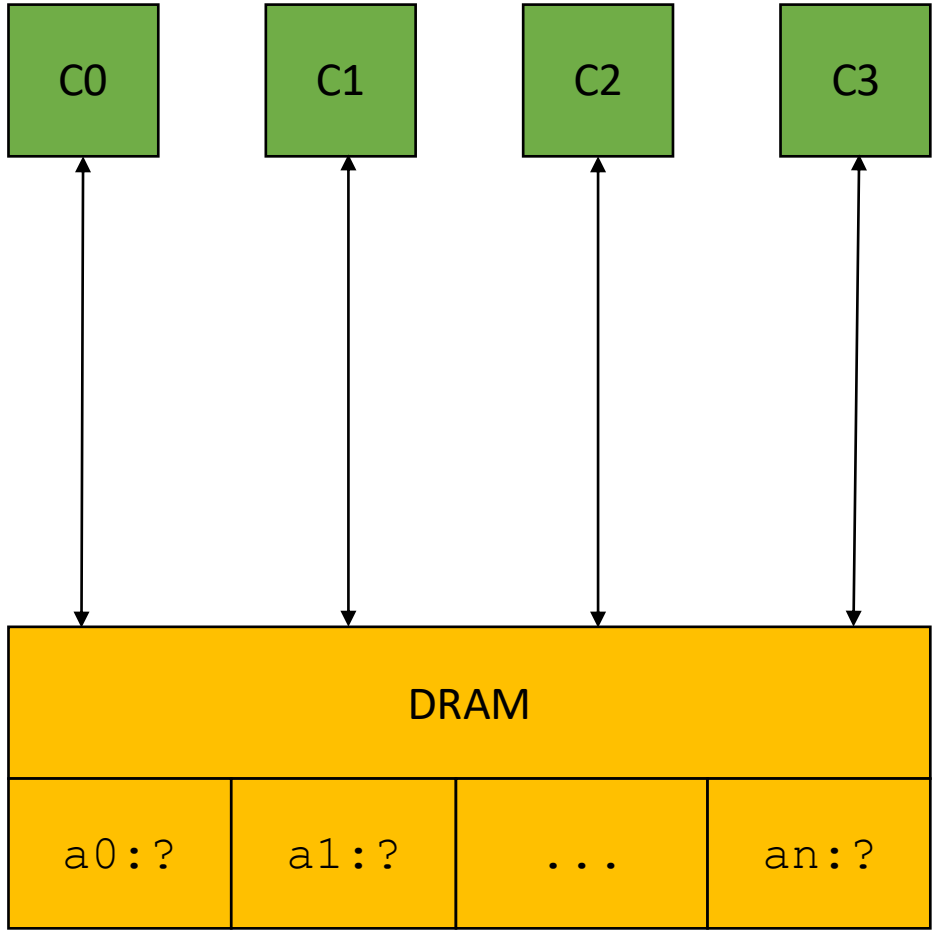
Core



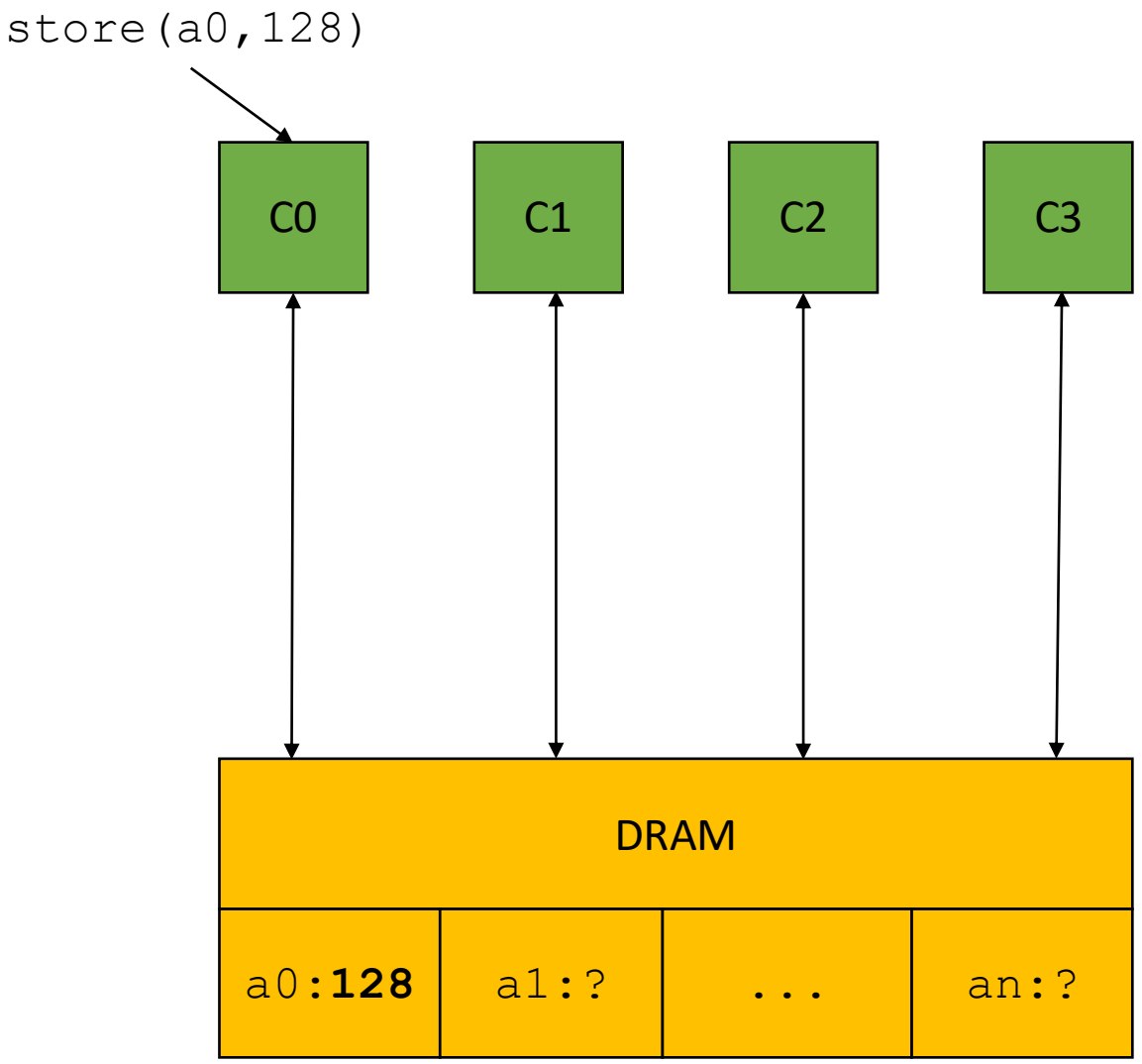
Core

Main memory

`store(a0,128)`

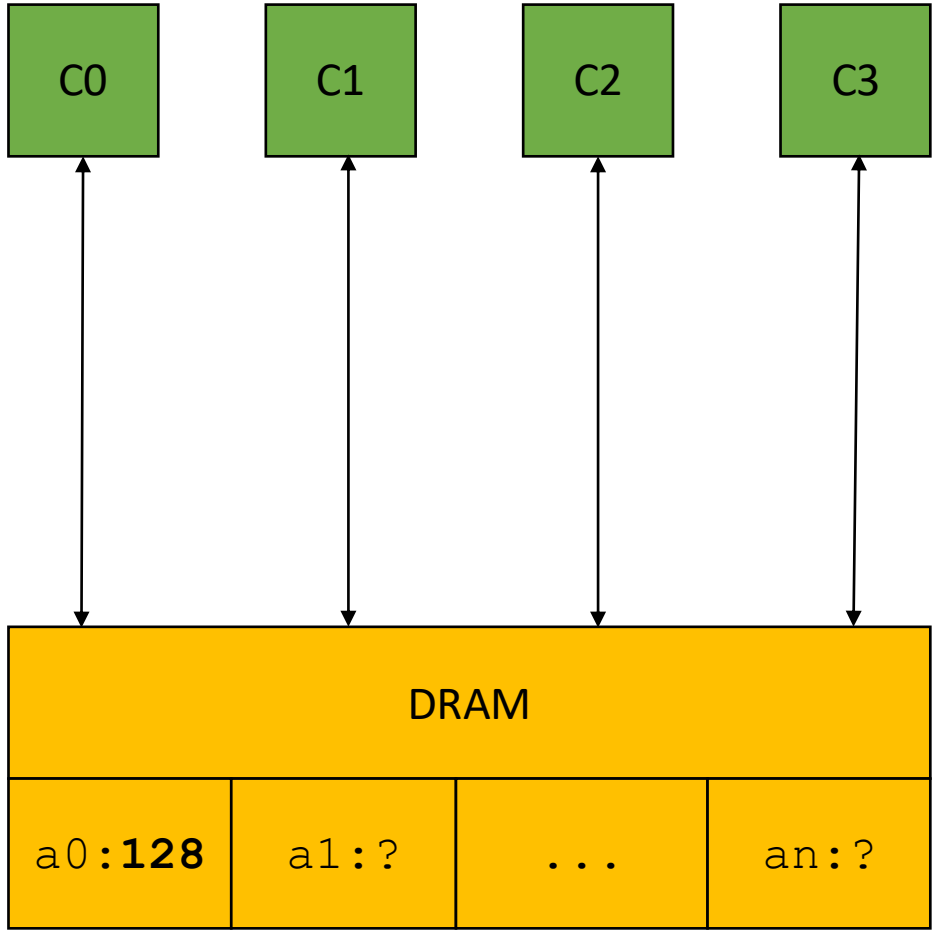


Main memory

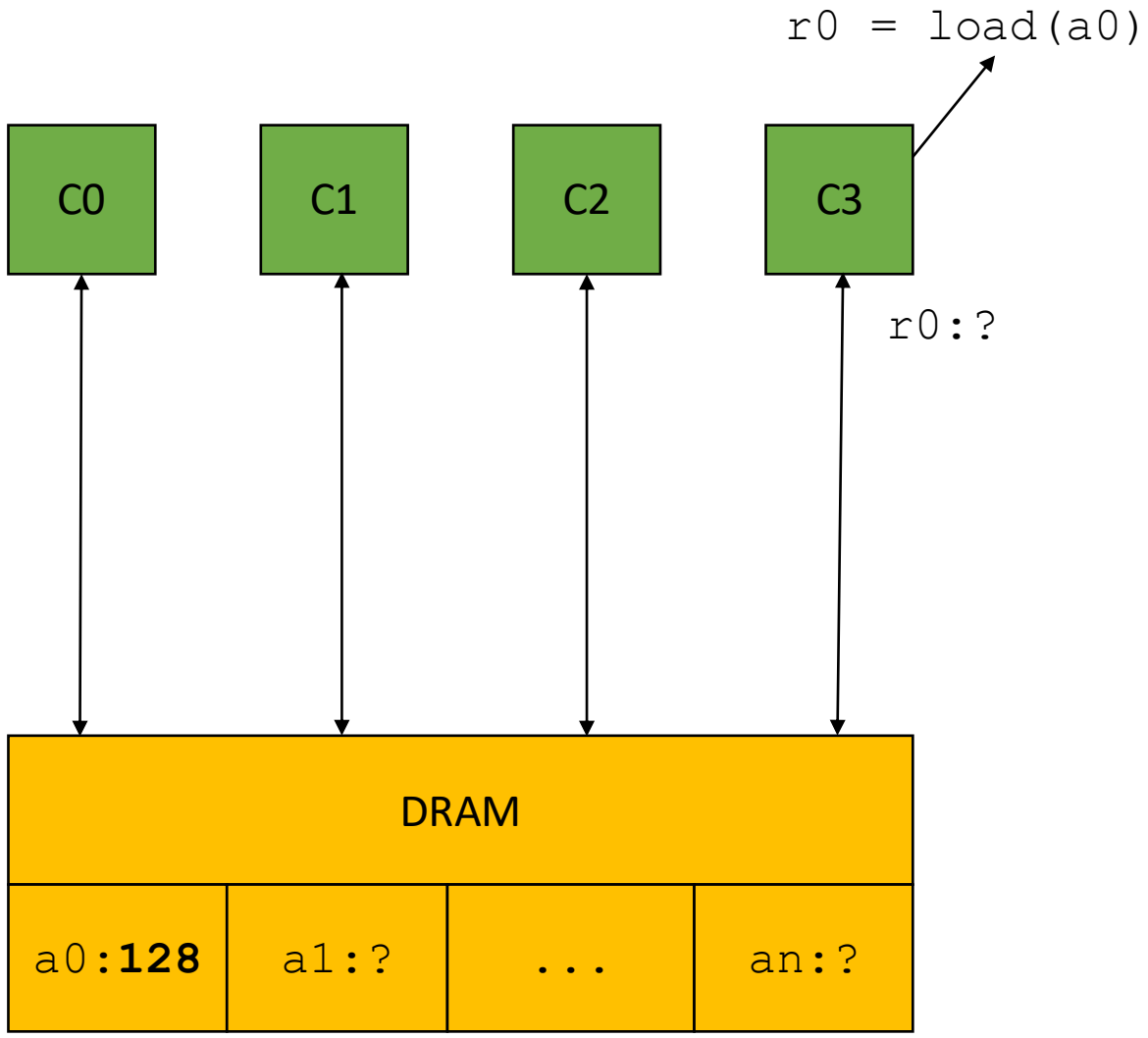


Main memory

```
r0 = load(a0)
```

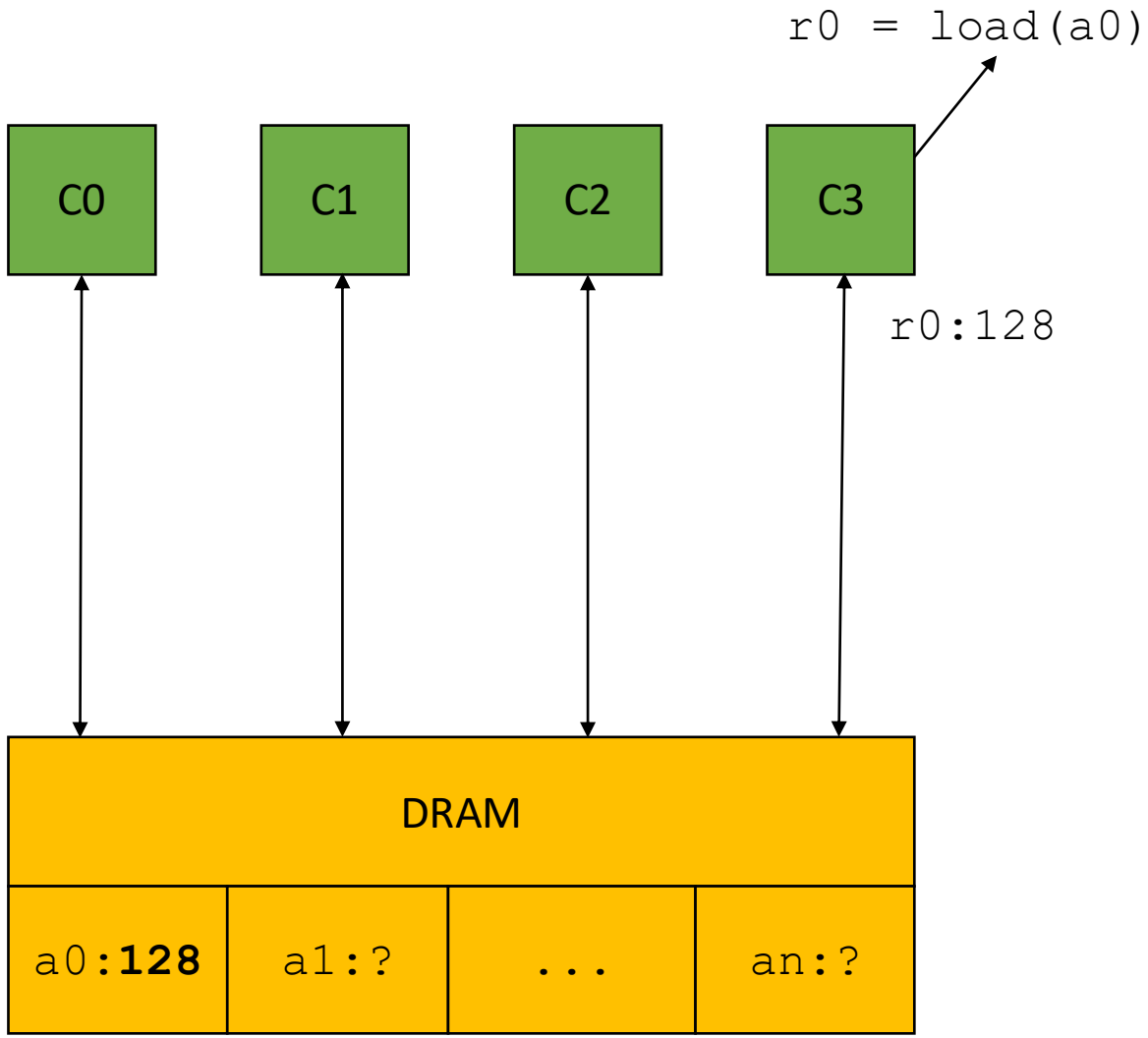


Main memory



Main memory

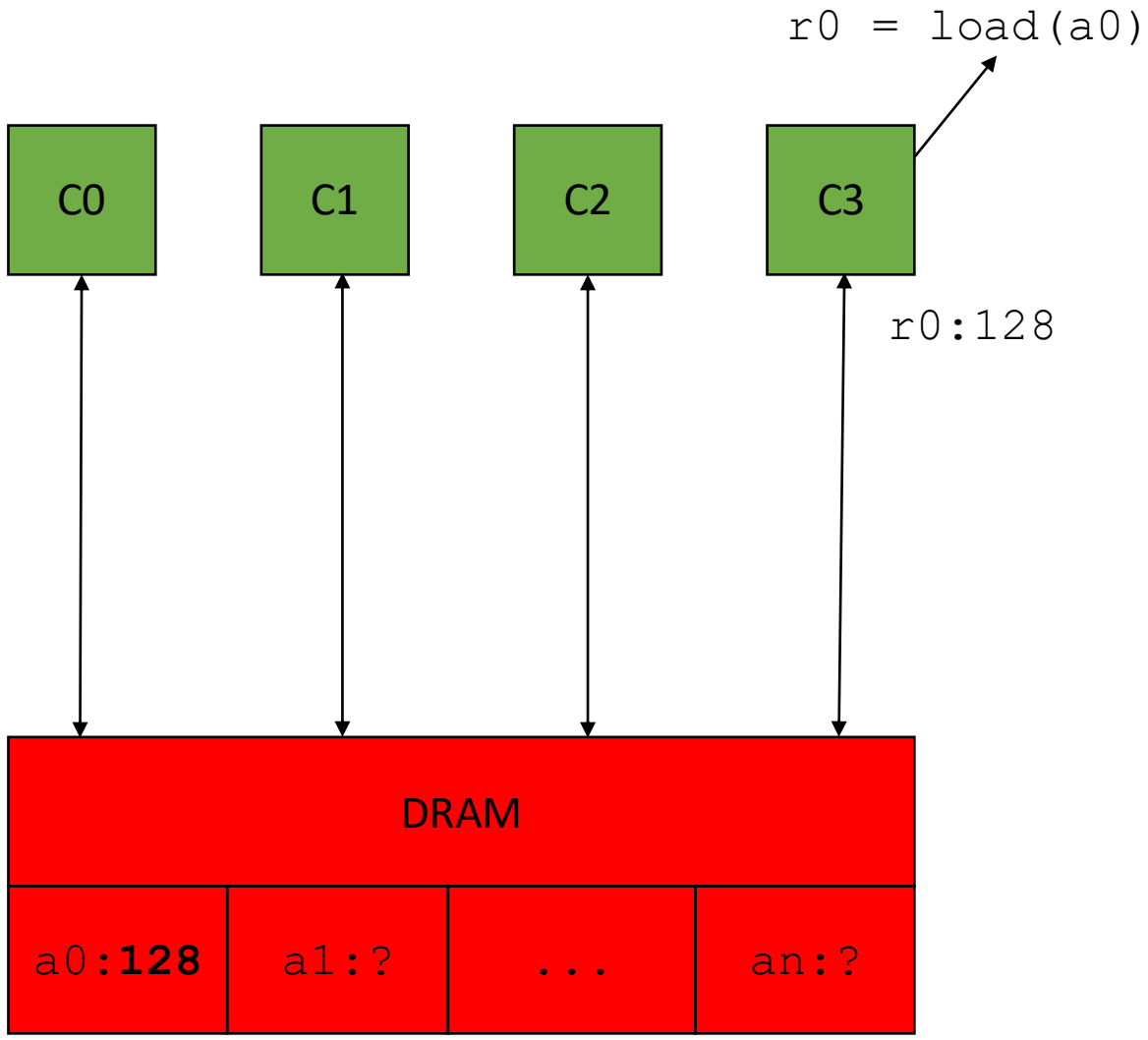
Problem solved!
Threads can communicate!



Main memory

Problem solved!
Threads can communicate!

reading a value takes ~200 cycles

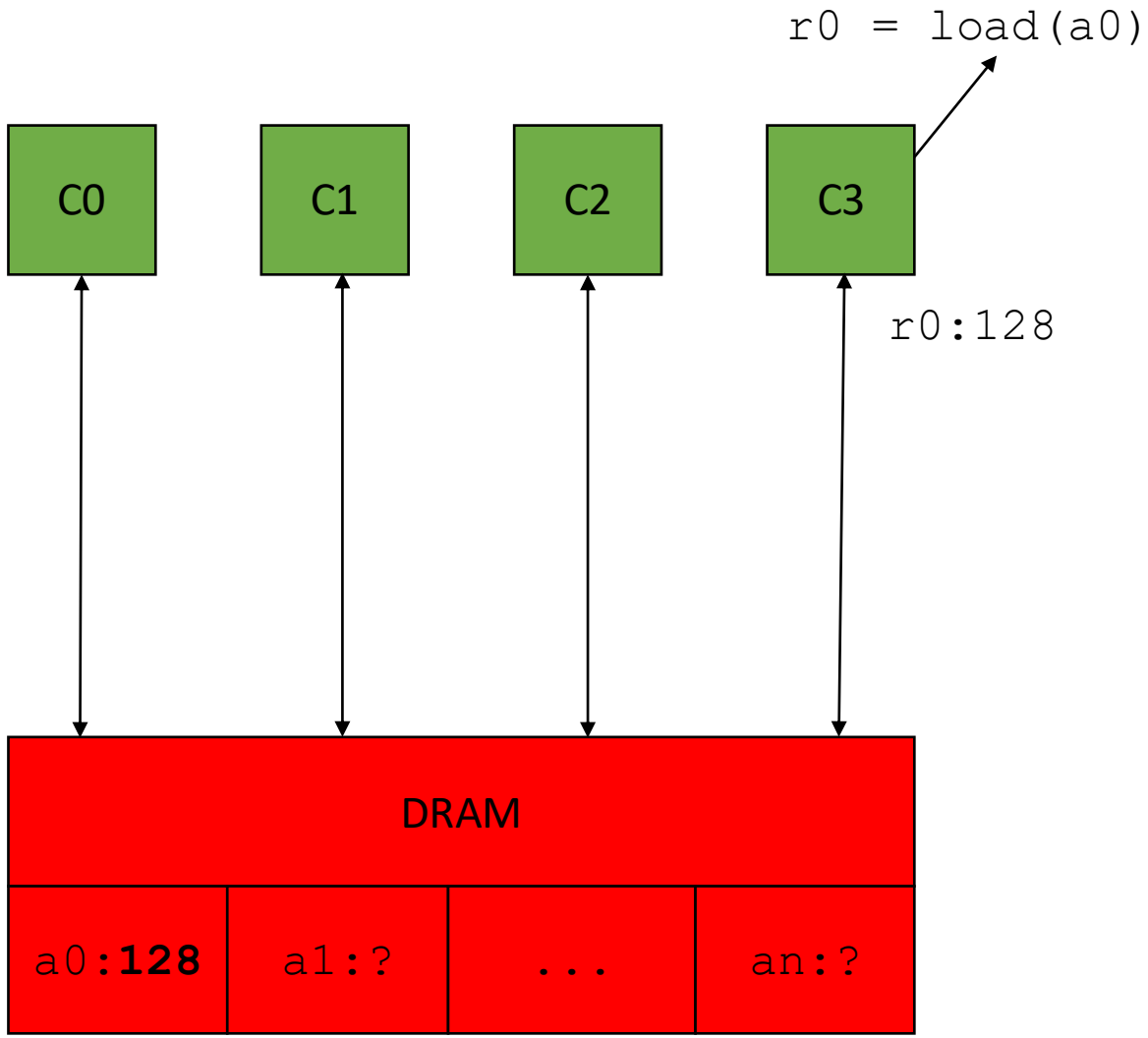


Main memory

Problem solved!
Threads can communicate!

reading a value takes ~200 cycles

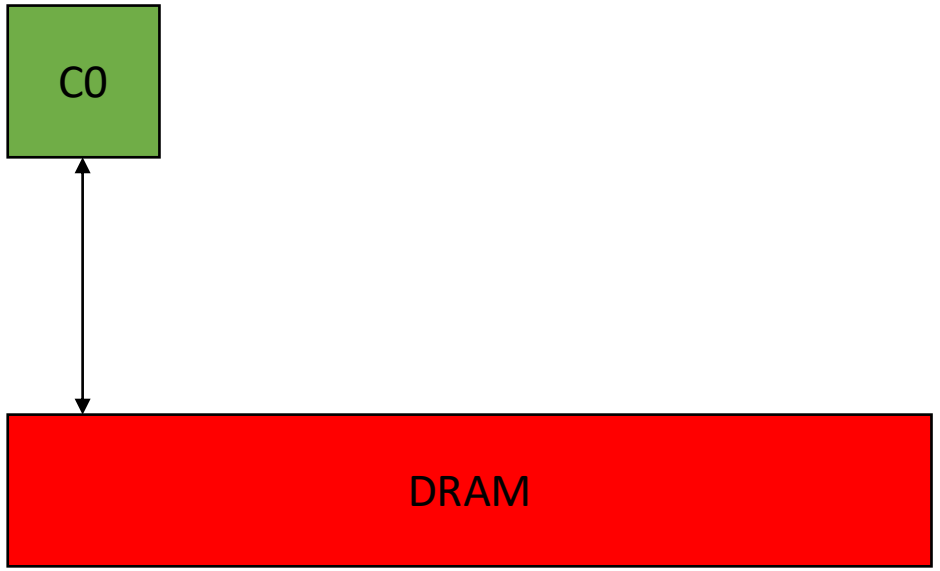
Bad for parallelism, but
also really bad for sequential
code (which we optimized for
decades!)



Main memory

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

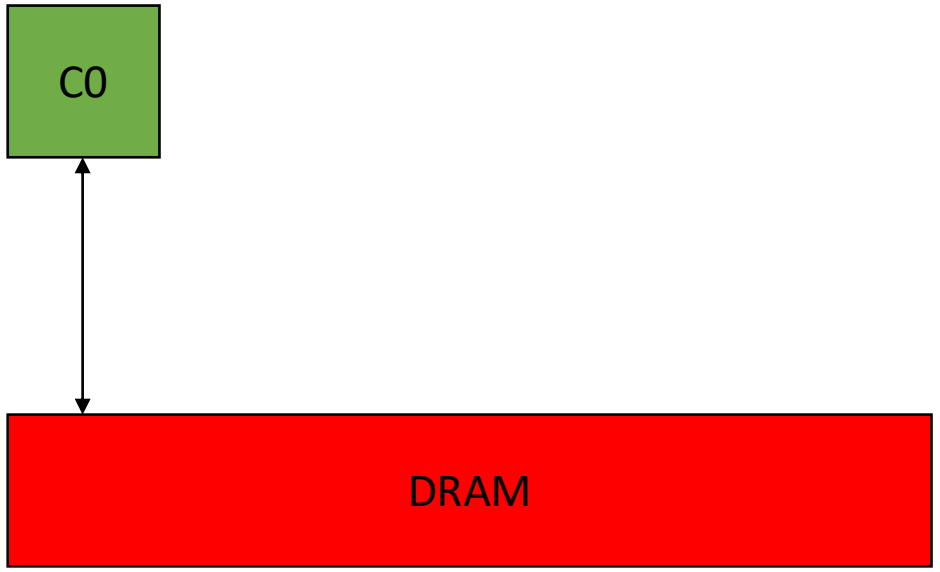


Main memory

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

200 cycles



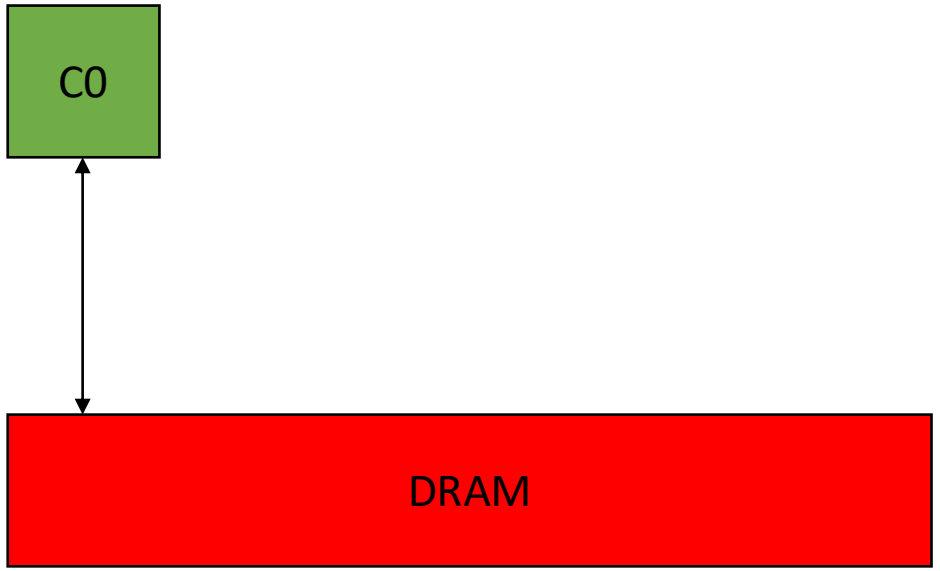
Main memory

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

200 cycles

1 cycles



Main memory

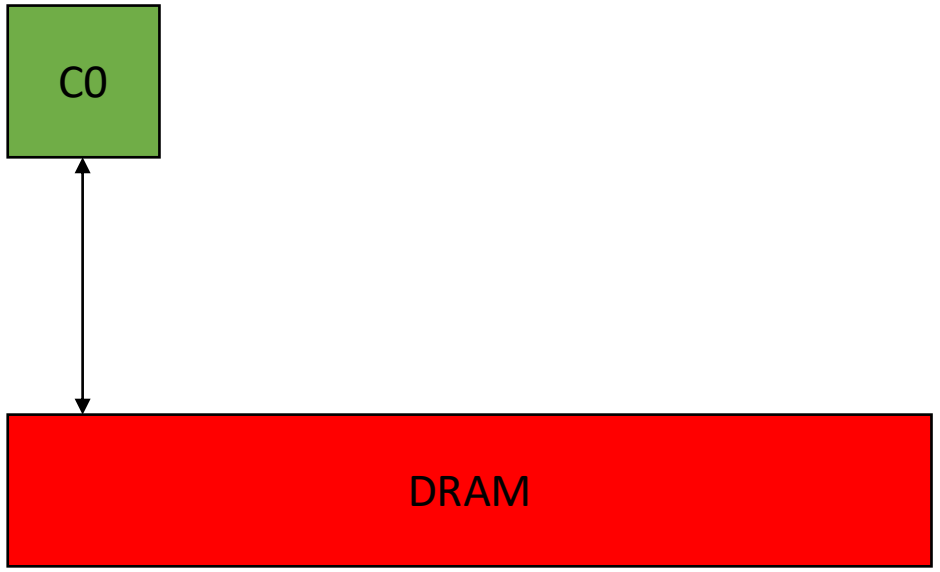
```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

200 cycles

1 cycles

200 cycles



Main memory

```
int increment(int *a) {
    a[0]++;
}
```

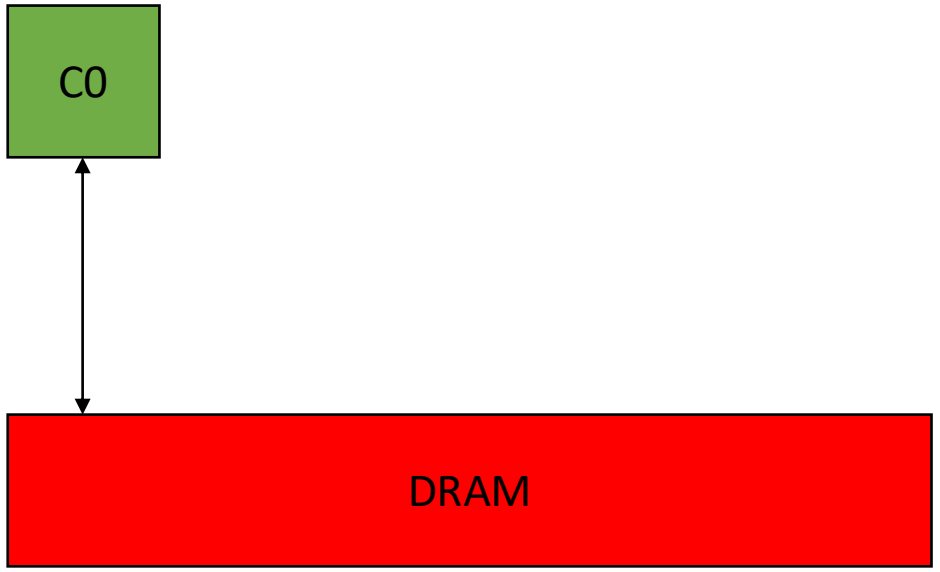
```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

200 cycles

1 cycles

200 cycles

401 cycles



Main memory

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

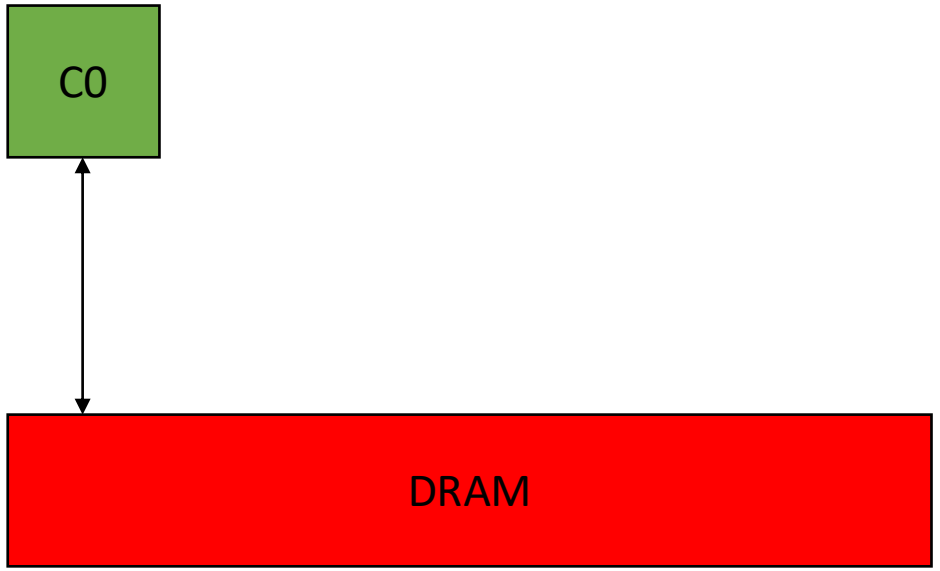
200 cycles

1 cycles

200 cycles

401 cycles

```
int x = 0;
for (int i = 0; i < 100; i++) {
    increment(&x);
}
```



Main memory

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

200 cycles

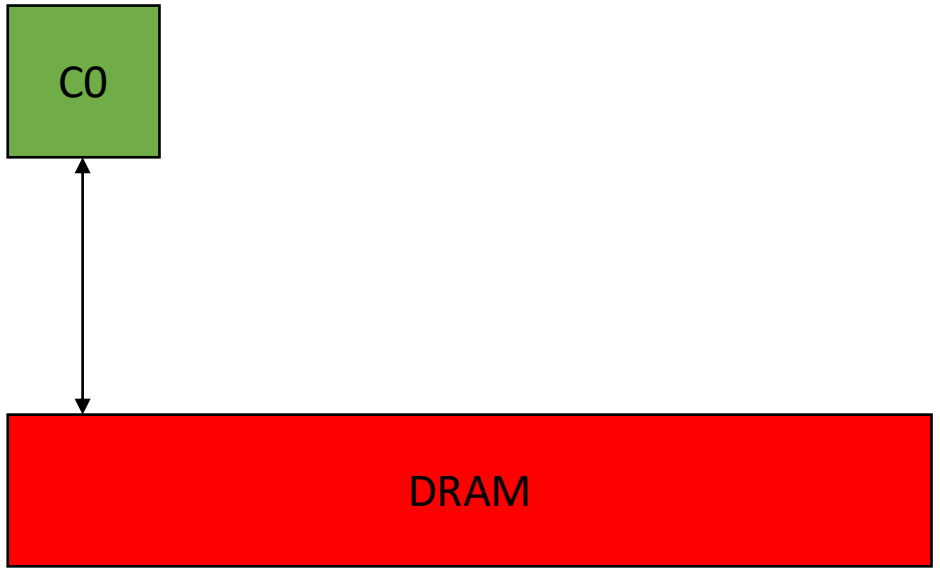
1 cycles

200 cycles

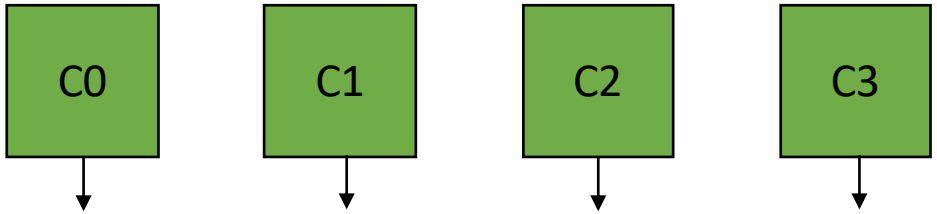
401 cycles

```
int x = 0;
for (int i = 0; i < 100; i++) {
    increment(&x);
}
```

40100 cycles!



Caches

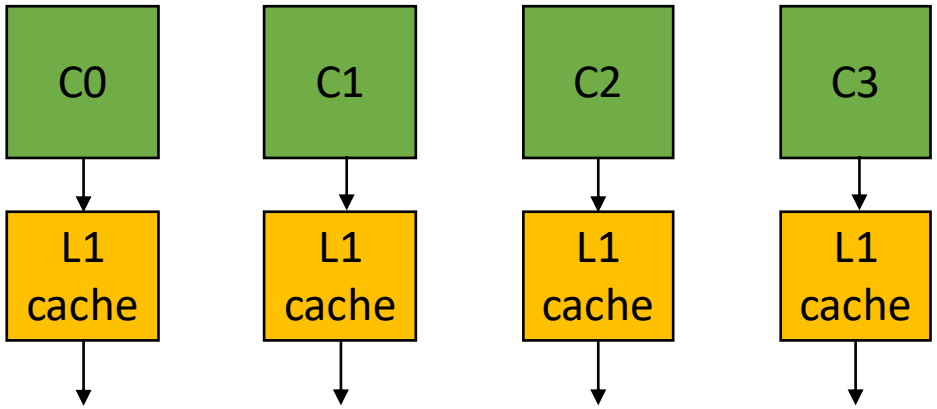


latency
~200 cycles



Many GBs
(or even TBs)

Caches



latency
~4 cycles

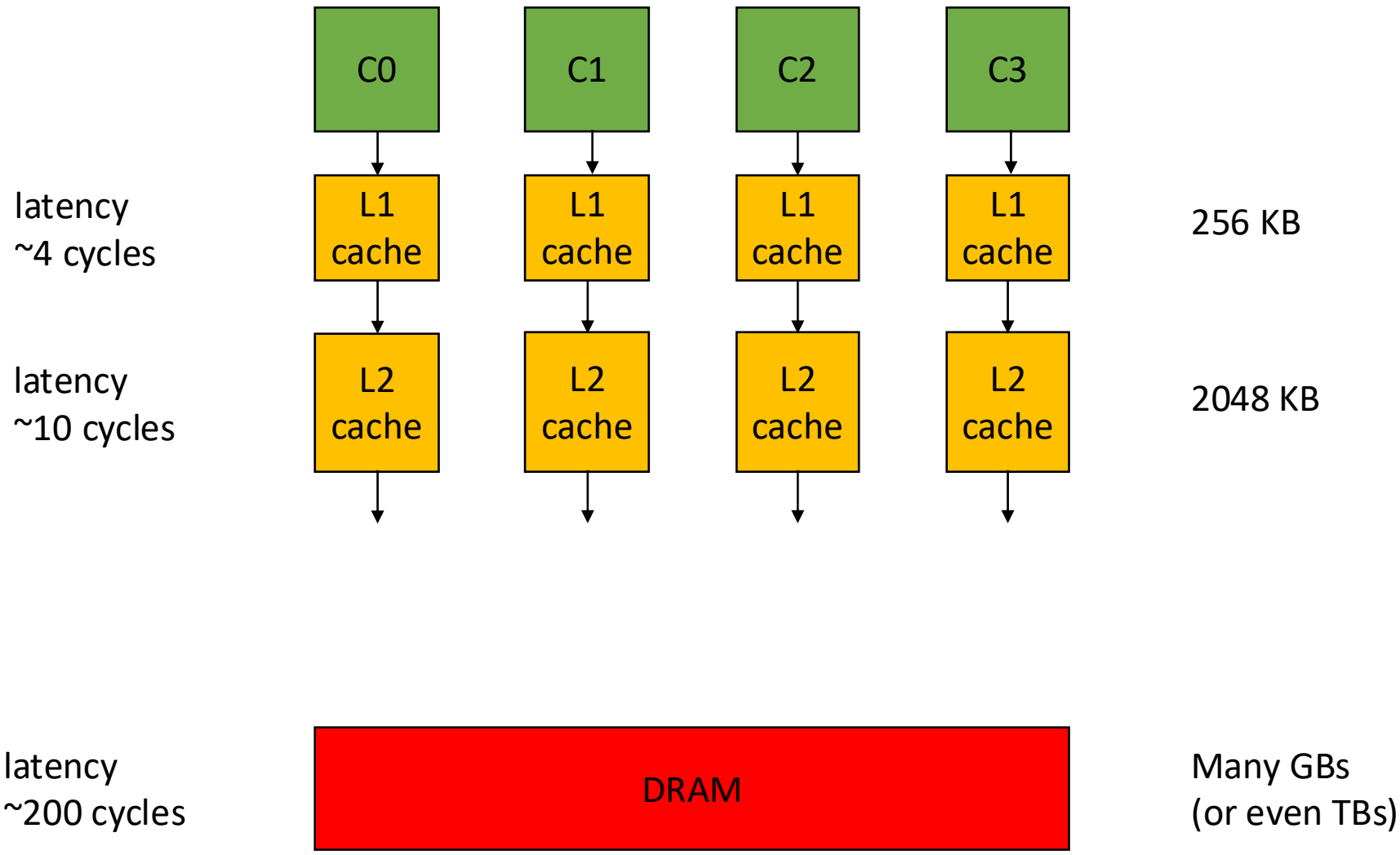
256 KB

latency
~200 cycles

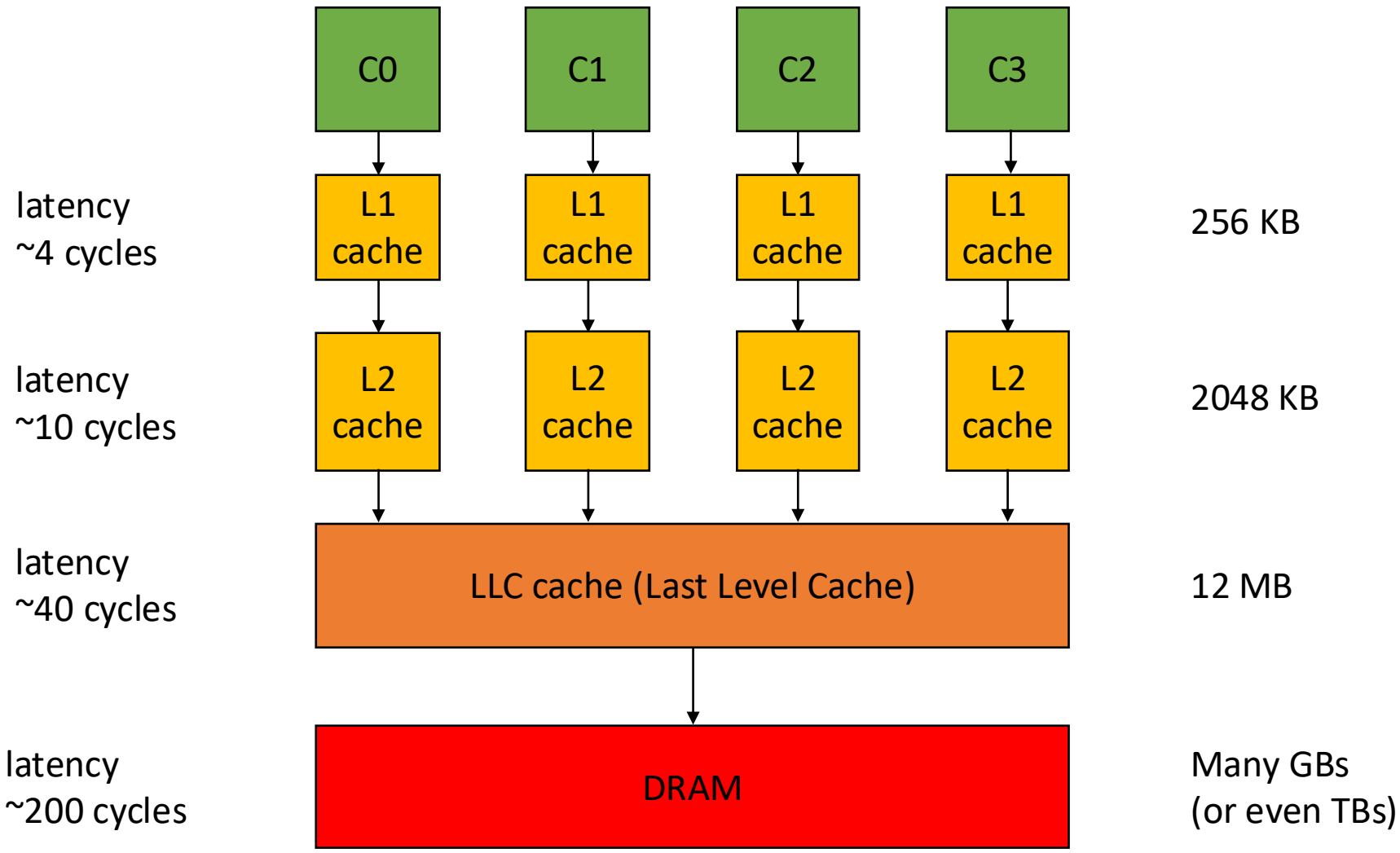
DRAM

Many GBs
(or even TBs)

Caches


C0
C1
C2
C3
latency
~4 cycles
L1
cache
L1
cache
L1
cache
L1
cache
256 KB
latency
~10 cycles
L2
cache
L2
cache
L2
cache
L2
cache
2048 KB
latency
~200 cycles
DRAM
Many GBs
(or even TBs)

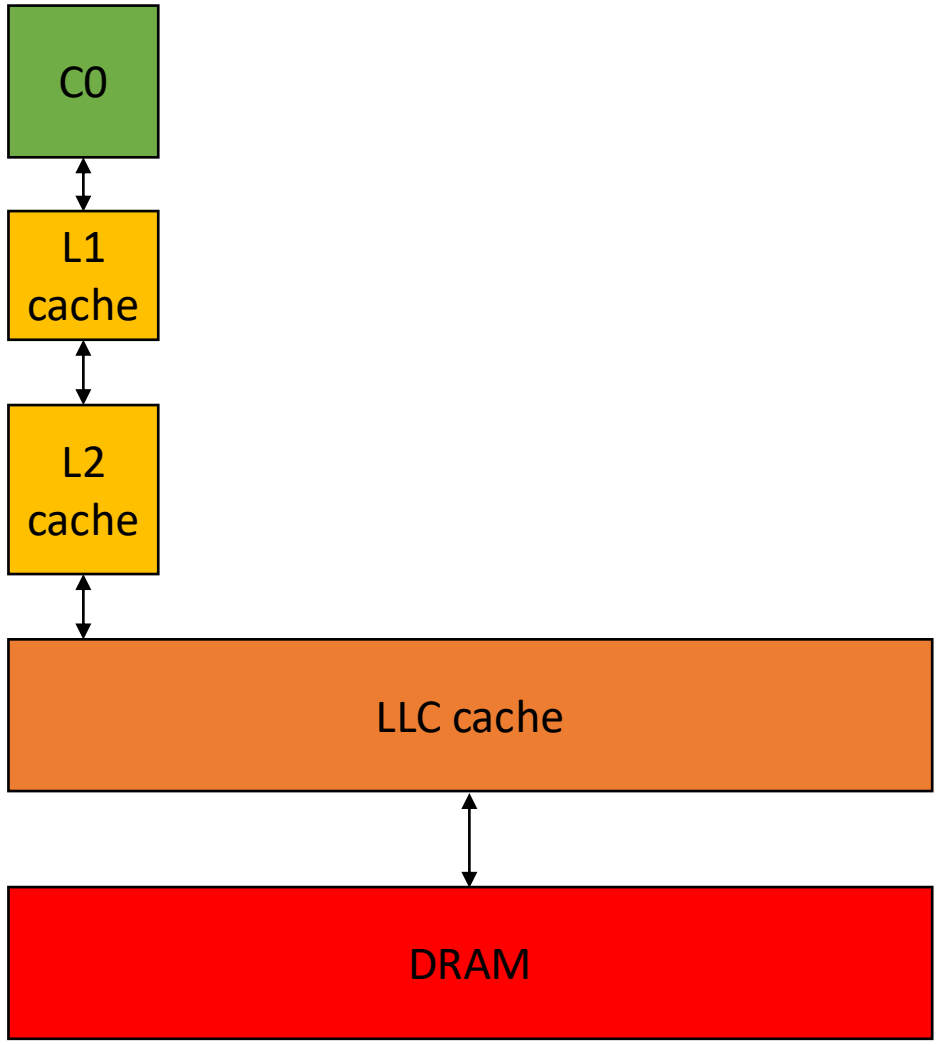
Caches


C0
C1
C2
C3
latency
~4 cycles
L1
cache
L1
cache
L1
cache
L1
cache
256 KB
latency
~10 cycles
L2
cache
L2
cache
L2
cache
L2
cache
2048 KB
latency
~40 cycles
LLC cache (Last Level Cache)
12 MB
latency
~200 cycles
DRAM
Many GBs
(or even TBs)

Caches

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```



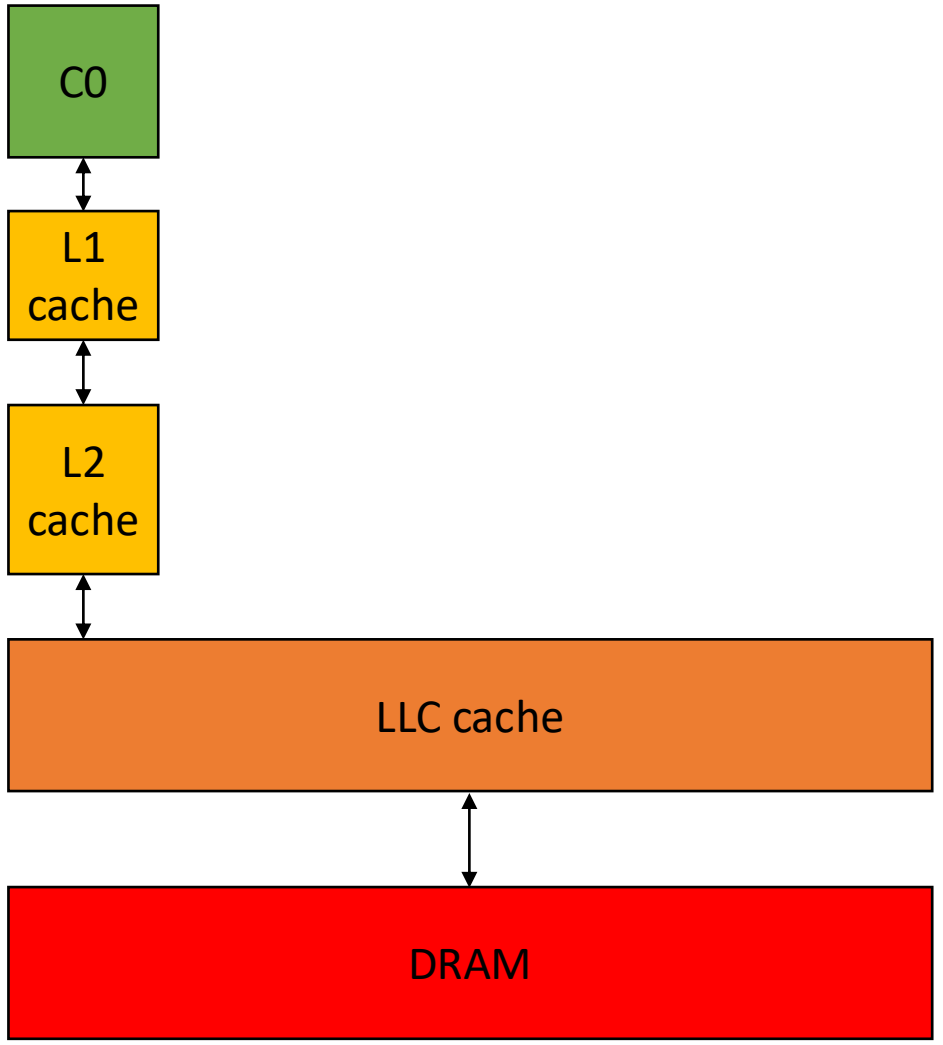
Caches

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

4 cycles

Assuming the value is in the cache!



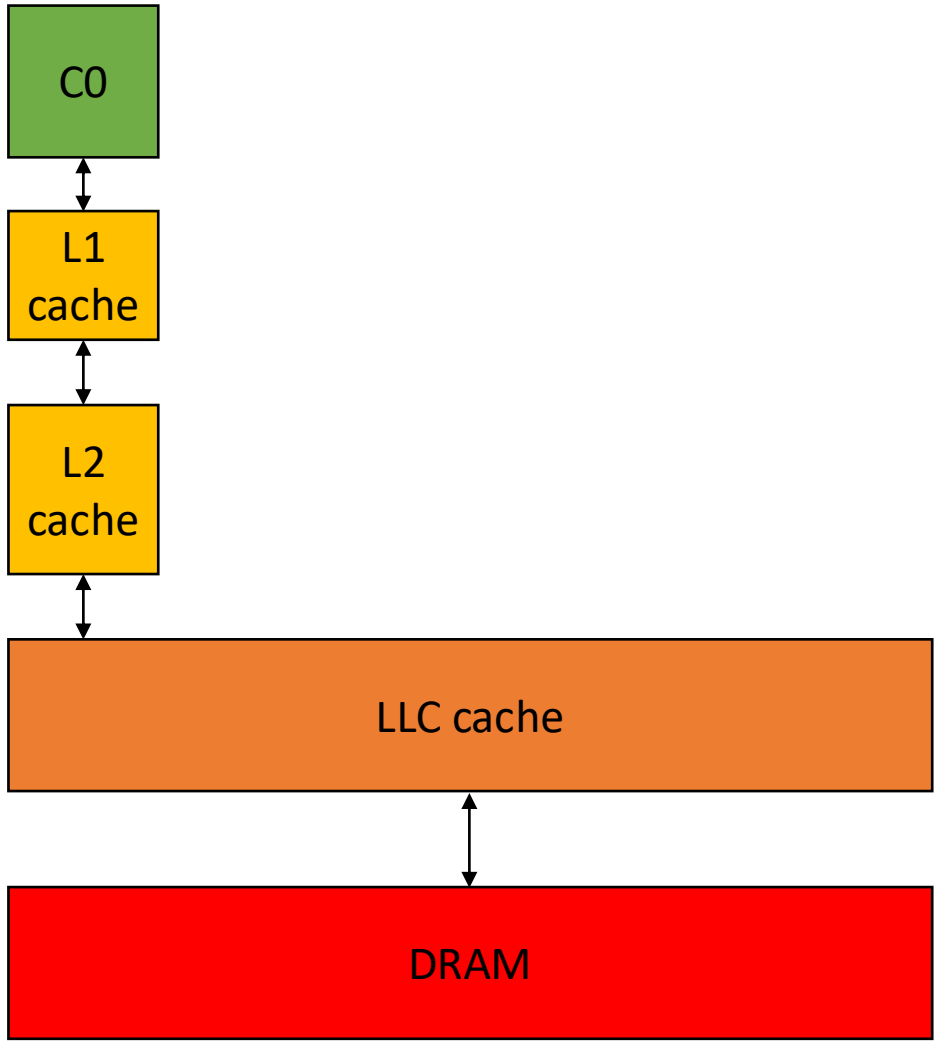
Caches

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

4 cycles

1 cycles



Caches

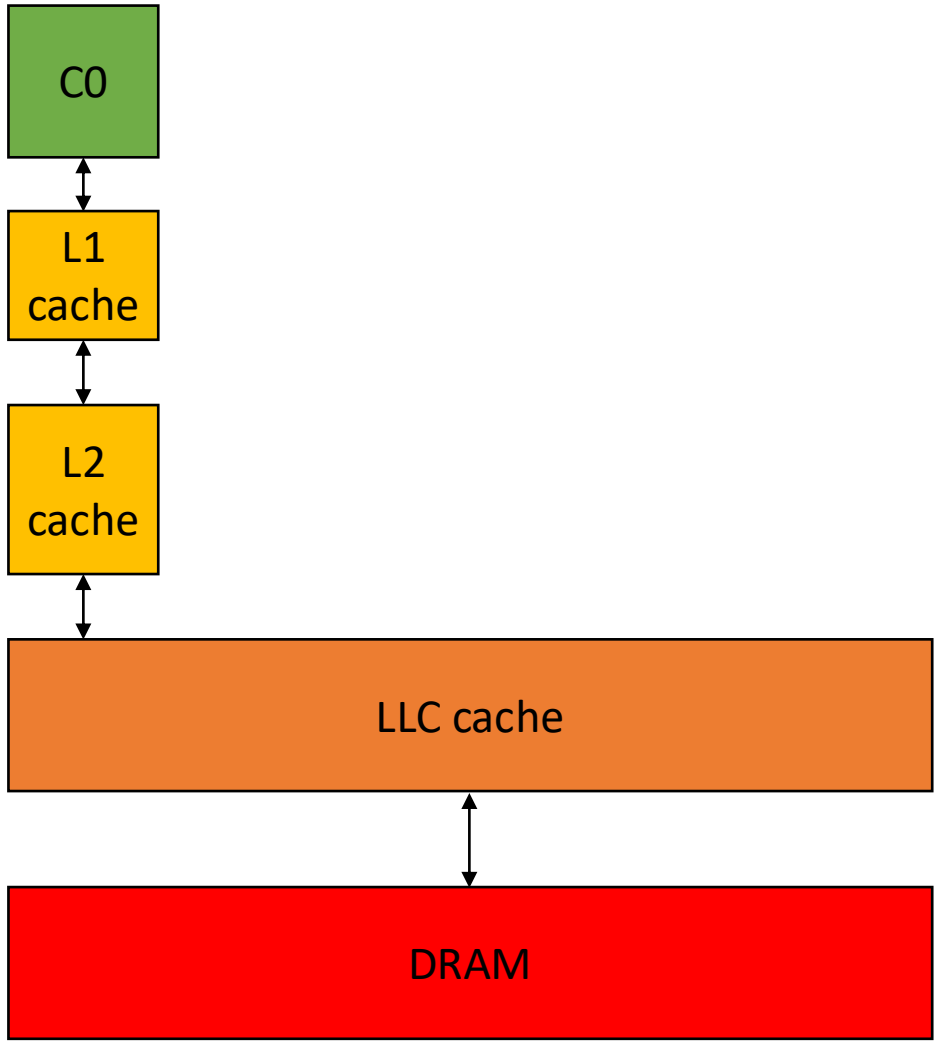
```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

4 cycles

1 cycles

4 cycles



Caches

```
int increment(int *a) {
    a[0]++;
}
```

```
%5 = load i32, i32* %4
%6 = add nsw i32 %5, 1
store i32 %6, i32* %4
```

4 cycles
1 cycles
4 cycles

9 cycles!

